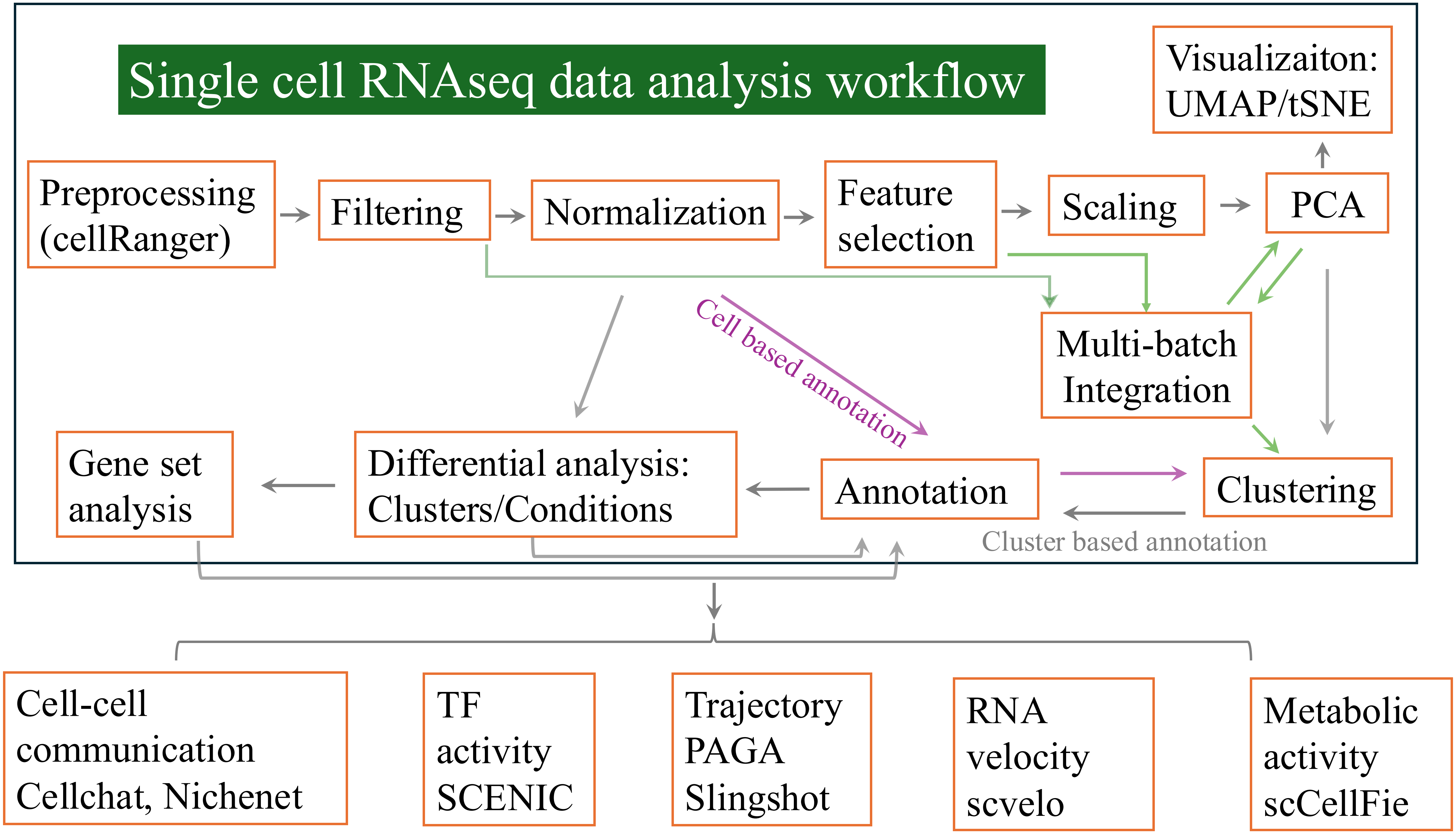
Basic 10X Genomics data analysis

Learnin goals

- Understand why filter data
- Understand why normalize data
- Understand why scale data?
- Understand why do dimensional reduction
- Know the popular dimensional reduction methods
- Understand why do clustering
- Understand why do differential analysis
- Understand why do cell annotation



Single cell RNAseq data analysis workflow
Visualizaiton: UMAP/tSNE
Preprocessing (cellRanger)
Filtering
Normalization
Feature selection
Scaling
PCA
Multi-batch Integration
Cell based annotation
Gene set analysis
Differential analysis: Clusters/Conditions
Annotation
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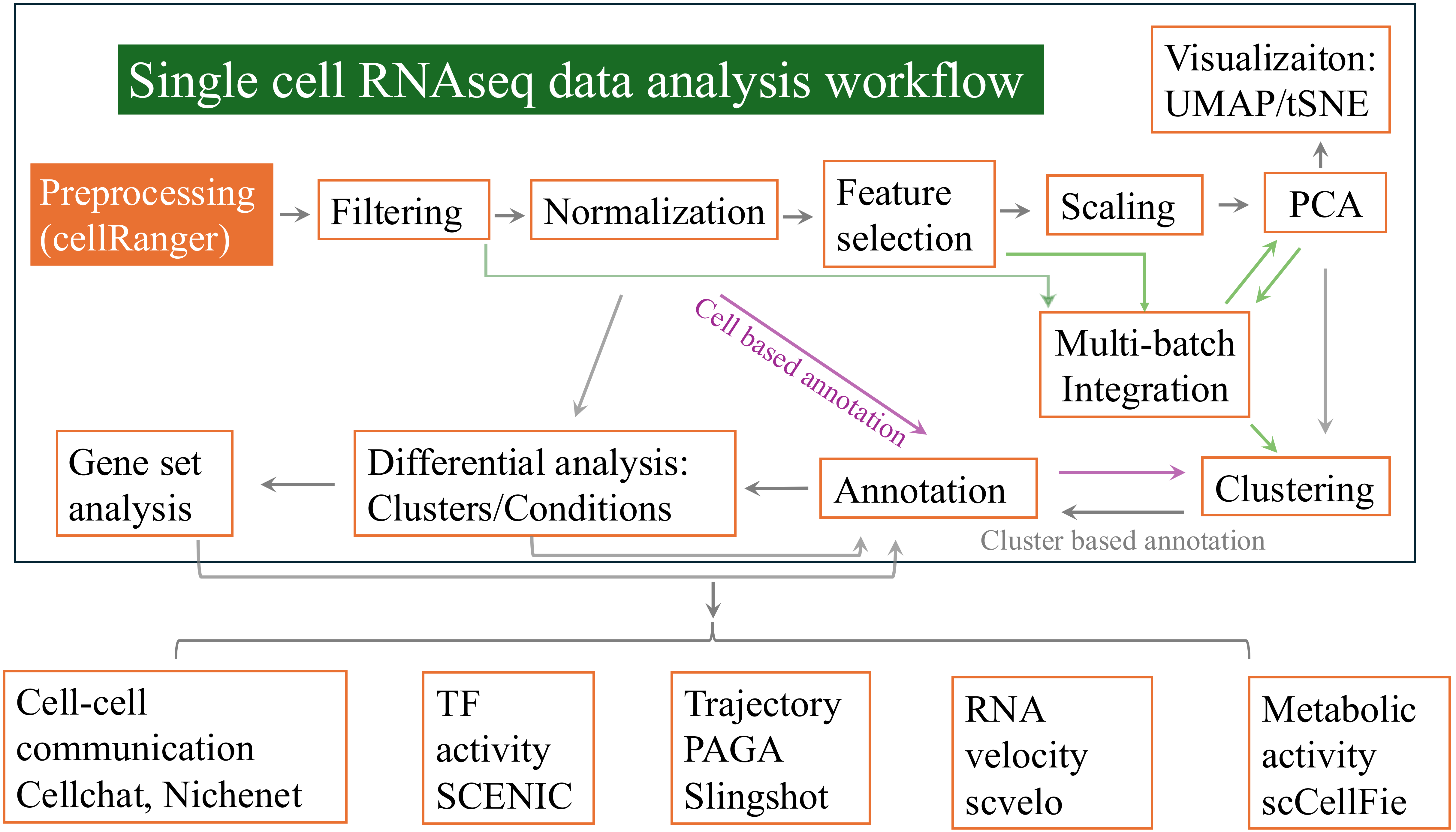
TF activity SCENIC
Trajectory PAGA Slingshot
RNA velocity scvelo
Metabolic activity scCellFie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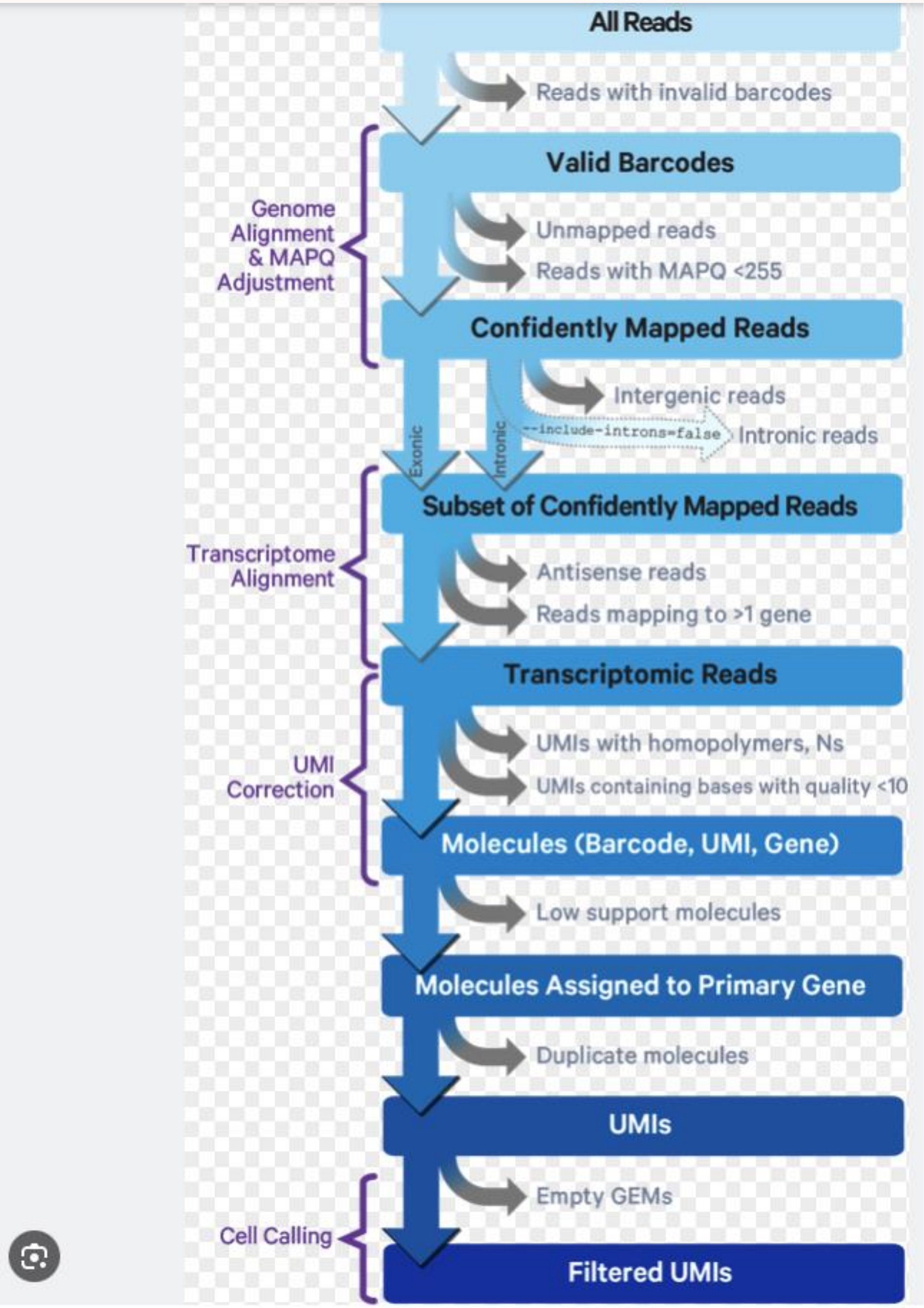
10X Genomics
Read mapping and counting

Cellranger Pipeline

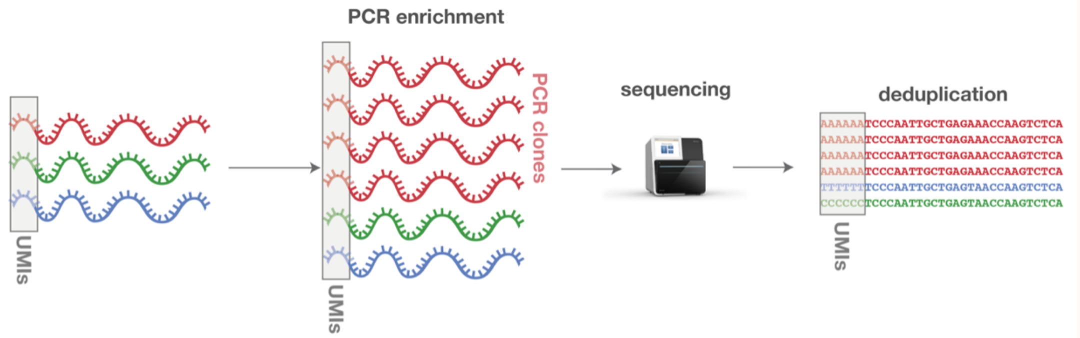
Get
gene x cell
count table



Cell Ranger's Gene Expression Algorithm - Official 10x Genomics Support

UMI: unique molecular identifiers

- Gene counts are UMI count
- Remove PCR introduced variation



https://www.ecseq.com

Count matrix

	Cell1	Cell2	..
Gene1	0	0	0
Gene2	0	1	0
Gene3	6	0	0
...	0	0	0

Question:

- **Single cell vs Bulk RNAseq:**
  - How they differ in mRNA quarntification (apart from cell vs tissue)

Downstream analysis_popular tools

- **Seurat** (R)
- Scanpy (python)
- Scran (R)

Data variation

Technical

- Cell quality (dying cells, broken cells…)
- Library prep efficiency (RNA capture, sequencing…)
- Batch effects
- …

Biological

- Cell cycle
- Sex, Age
- Transcriptional bursting
- Cell size
- Cell type/state
- …

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Count matrix

	Cell1	Cell2	..
Gene1	0	0	0
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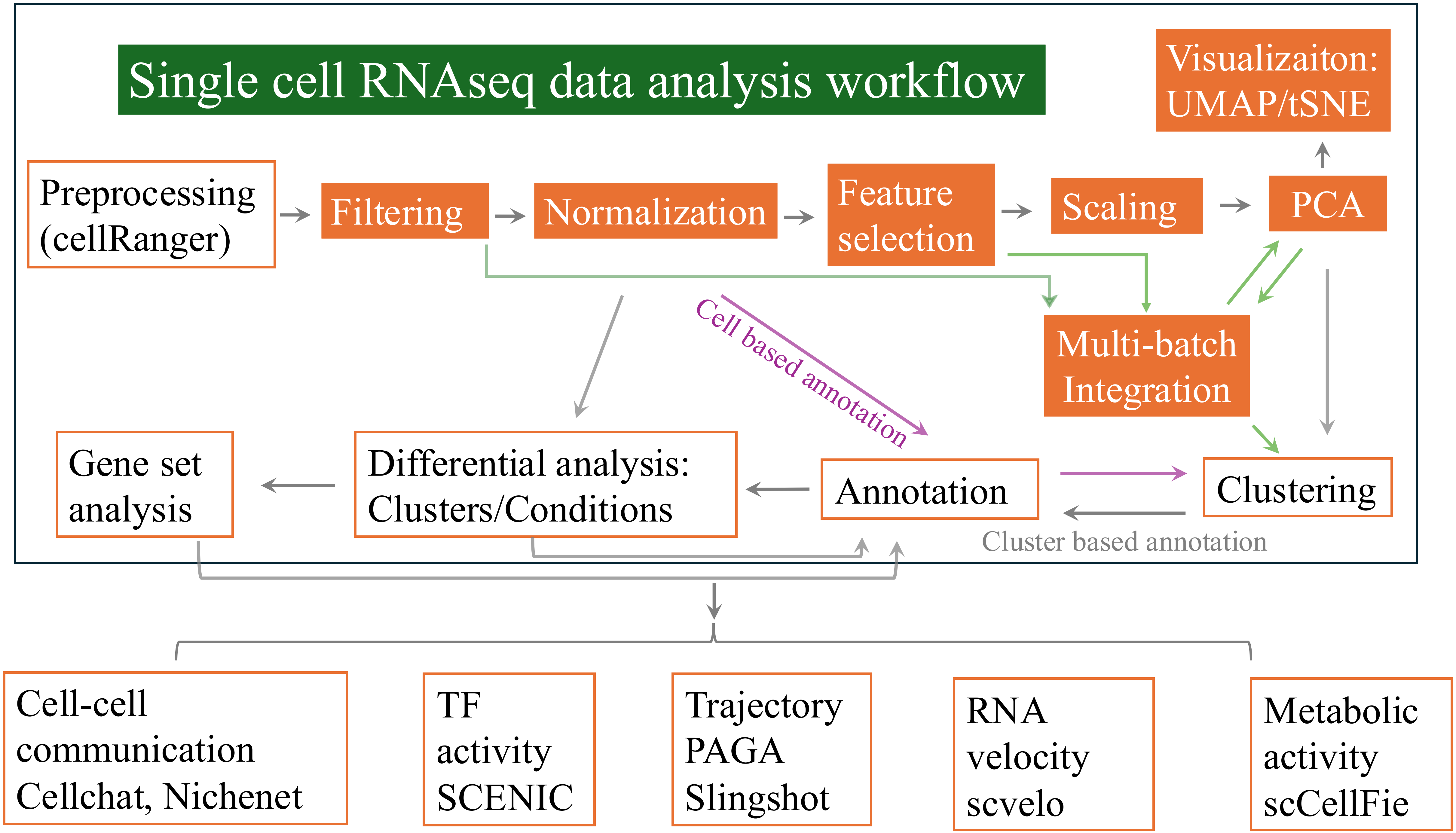
Gene3	6	0	0
...	0	0	0

Remove unwanted variation
to look for biological celltypes/states



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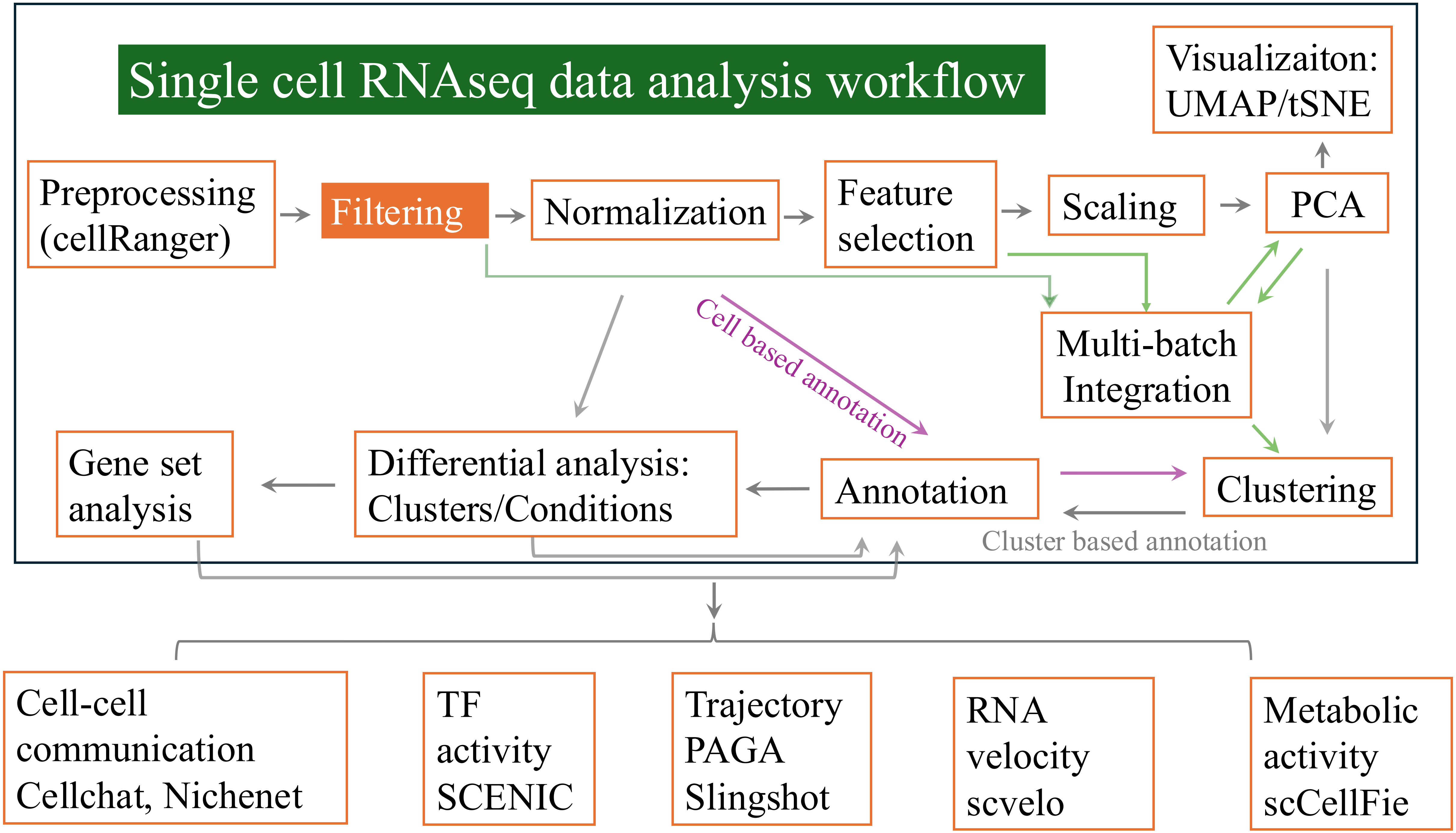
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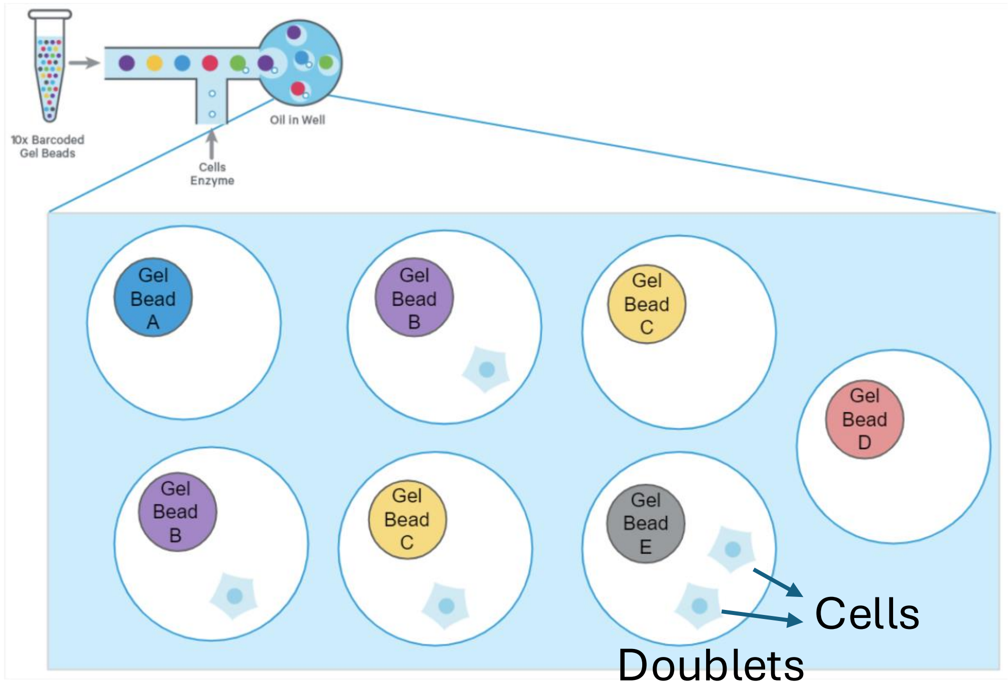
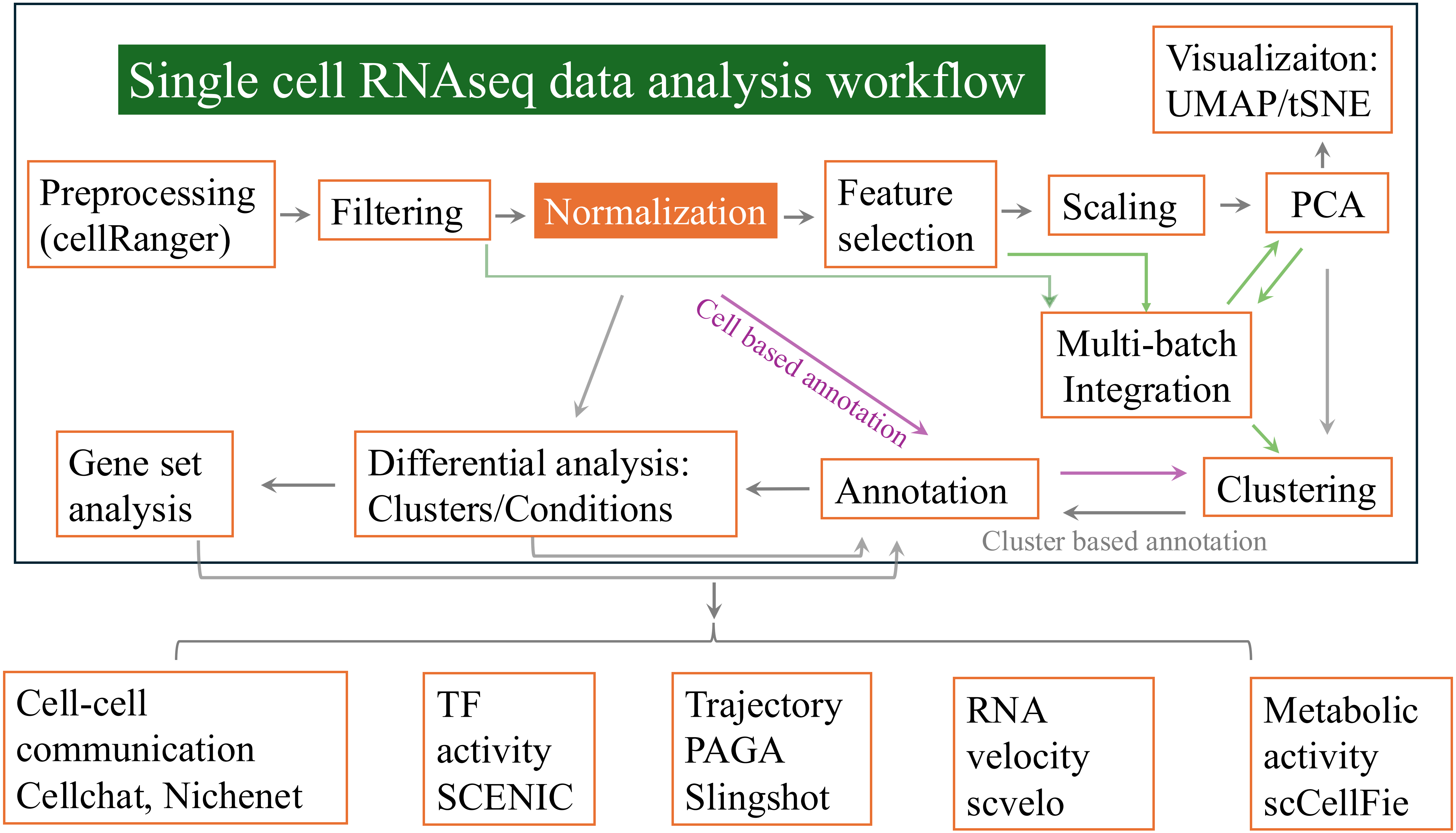
Filtering

- ❖Filter out genes
  - ➢Lowly captured genes (e.g. <=**3**cells)
- ❖Filter out RNAs
  - ➢**Cell-free, ´ambient´ RNA within cell suspension**
- ❖Only keep viable cells
  - ➢**Empty droplets** (few genes/UMI)
  - ➢**Low viable cells** (few genes/UMI)
  - ➢**Doublets (too many genes/UMI)**
  - ➢**Broken cells** (leaking)
    - High mitochondrial gene percentage & few genes/UMI
    - Cytosolic trancripts easier to leak out than mitochondria (big)

Doublet: a 10X droplet capturing 2 cells



https://kb.10xgenomics.com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Normalization

Sampling effect in
- RNA capture by beads
- RNA —> cDNA
- Sequencing of cDNA

Cell cycle
Cell size
Transcriptional bust
Batch effect

⇨ Total cell UMI count differ among cells

Hafemeister & Satija 2019 20:296

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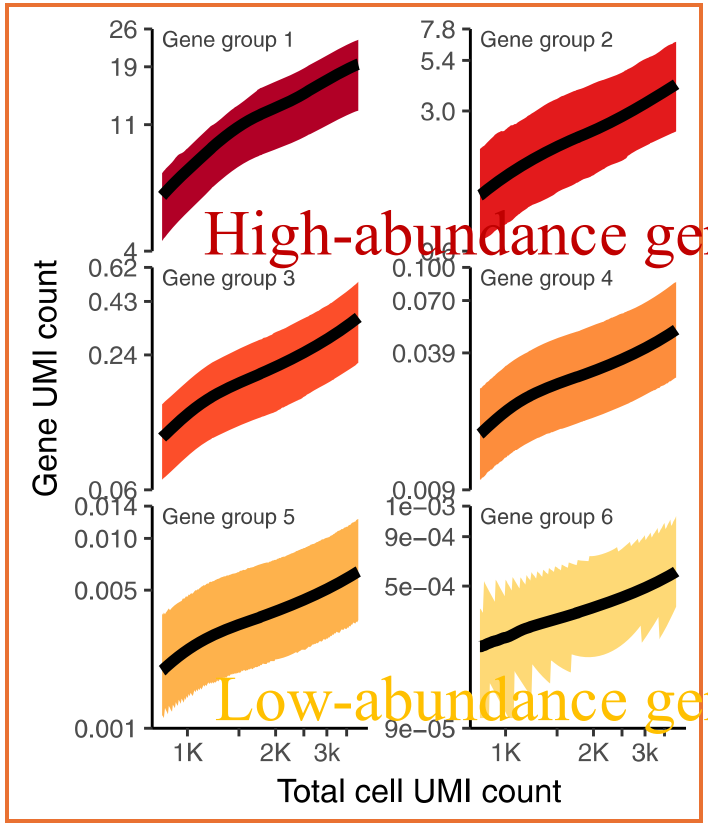

Hafemeister & Satija 2019 20:296

➢ Gene abundance depends on total cell UMI count **(desired biological differences?)**

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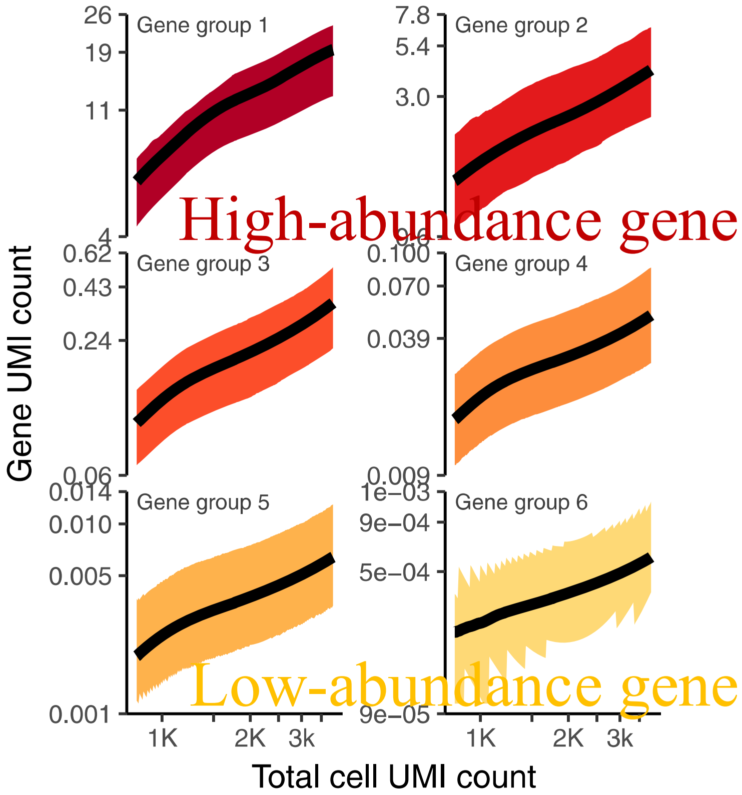


Hafemeister & Satija 2019 20:296

➢Gene abundance depends on total cell UMI count **(desired biological differences?)**

➢ **Normalization**
- (try to) Remove total UMI count differences among cells
- Make gene expression comparable between cells

Normalization methods

➢ Seurat: LogNormalize (Scaling)

* *Assume all cells with the same molecular composition*

$$X_i = \frac{\text{The read count of gene X in cell } i}{\text{Total counts of cell } i} \times 10^4 \ (Eq.1)$$

$$f(x_i) = ln(x_i + 1) \ (Eq.2)$$

More or less

RNA:count slot: raw count

Seurat::NormalizeData()

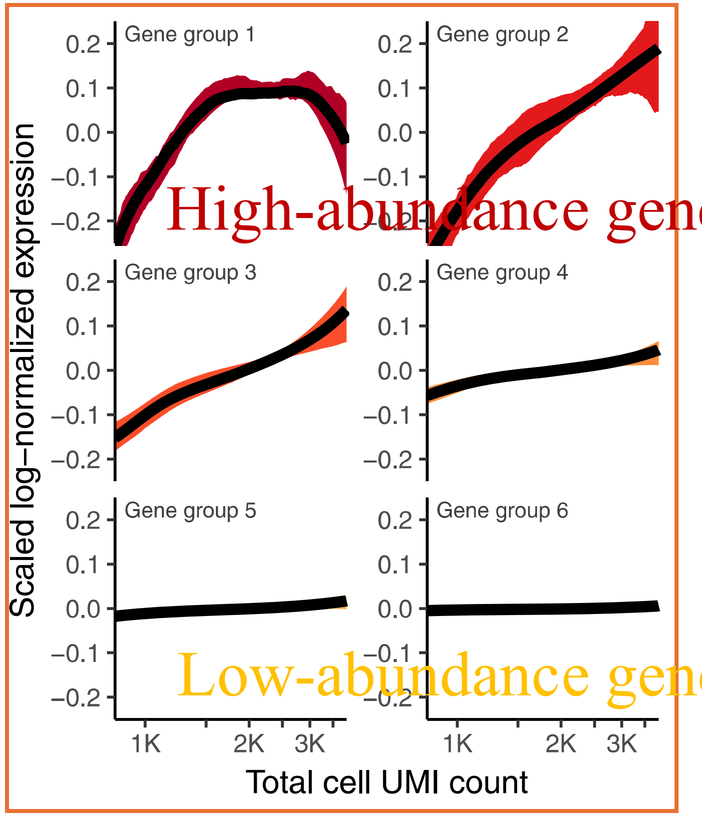
Each cell's counts are:

1. **Divided by the total counts** for that cell (to no
2. **Multiplied by a scale factor** (default = 10,000)
3. **Log-transformed** (using natural log).

$$x'_{ij} = \log\left(\frac{x_{ij}}{\text{total}_j} \times 10{,}000 + 1\right)$$

- x_{ij}: raw count of gene *i* in cell *j*
- total_j: total counts in cell *j*

RNA:data slot:
* Normalized count
* Near normal distribution
* All cells the same total UMIs



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Normalization methods

- Seurat: LogNormalize (Scaling)
  * Assume all cells with the same molecular
composition
- High-count filtering CPM
- BASiCs (Spike-in based)
- Scran (Lun et al. 2016) (Pooling-based)
- **Downsampling**

Normalization methods

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- High-count filtering CPM
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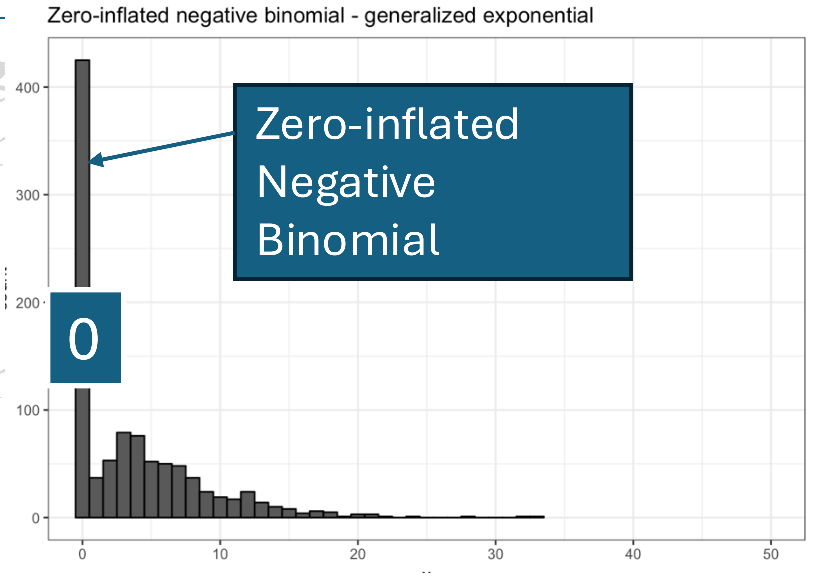
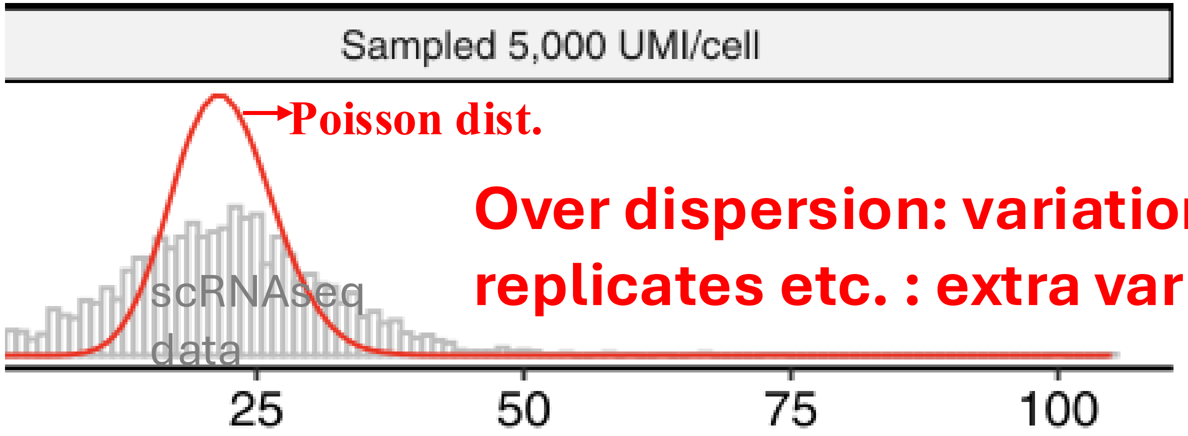
- Scran (Lun et al. 2016) (Poolir
- Downsampling
- ZINB-Wave (overfitting)
- **Regularized Negative Binomial Regression/Seurat: SCTransform v2**



Model parameters may differs between datasets: normalize separately

Single cell RNAseq data modeling:

❖ **Count: poisson distribution (variation=mean)**

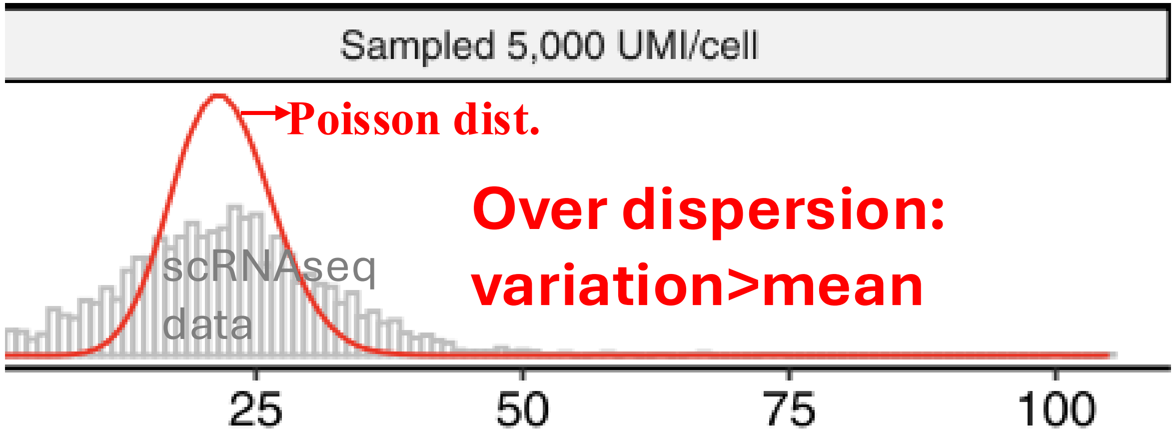


Over dispersion: variation>mean

replicates etc. : extra variation to count data

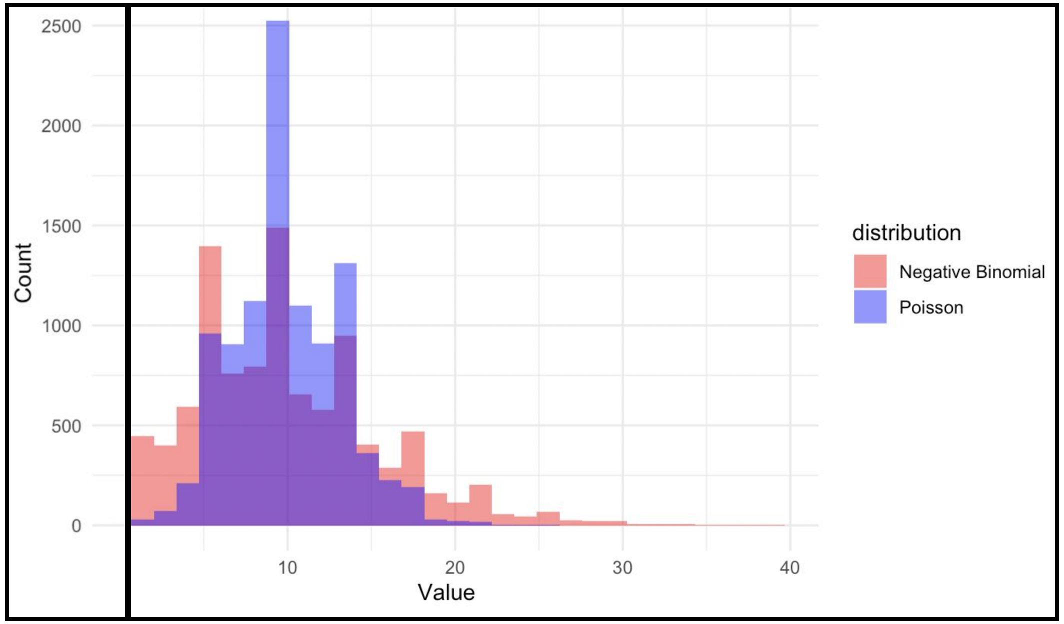
Single cell RNAseq data modeling:

❖ **Poisson distribution**



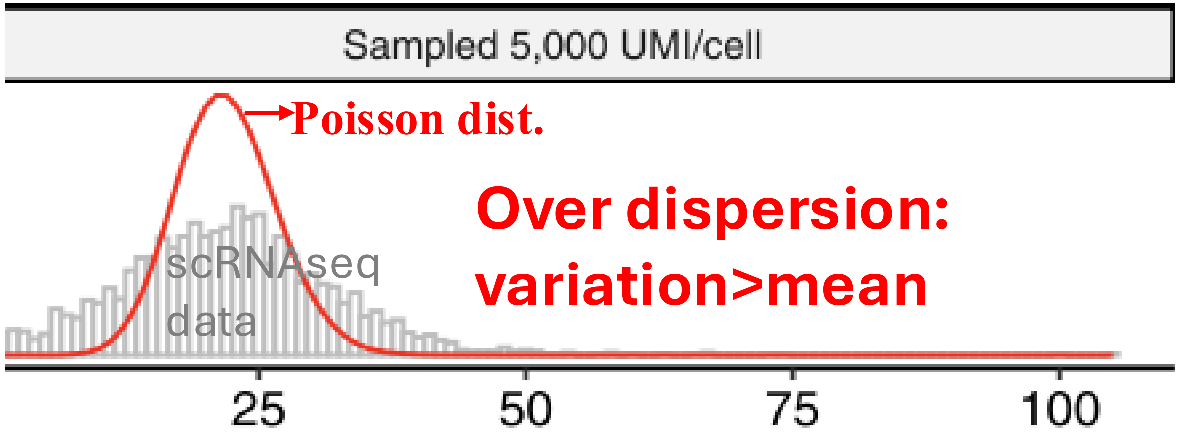
Over dispersion: variation>mean

❖ **Negative Bionomial**



Single cell RNAseq data modeling:

❖ Poisson distribution



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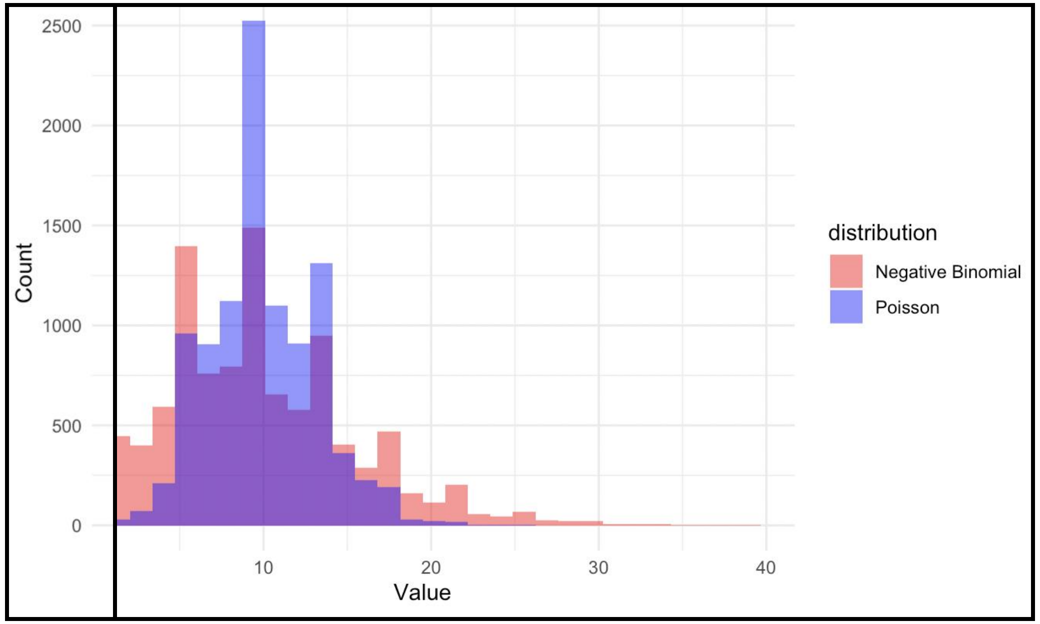


NB GLM:

Total cell UMI

$$\log(\mathbb{E}(x_i)) = \beta_0 + \boxed{\beta_1 \log_{10} m_i}$$

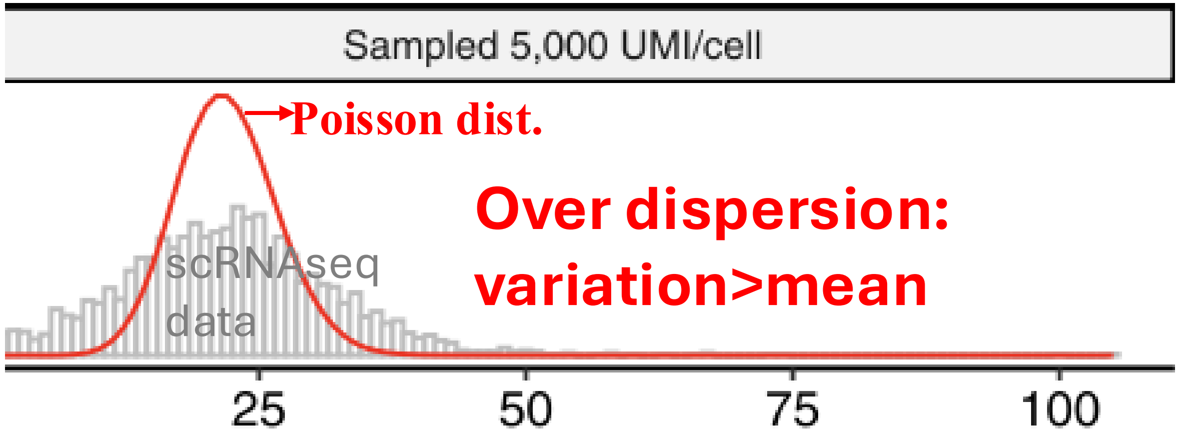
$y_i = NB(E(x_i), \theta)$

Dispersion parameter

Remove total cell UMI dependence

Single cell RNAseq data modeling:

❖ Poisson distribution (variation=mean)



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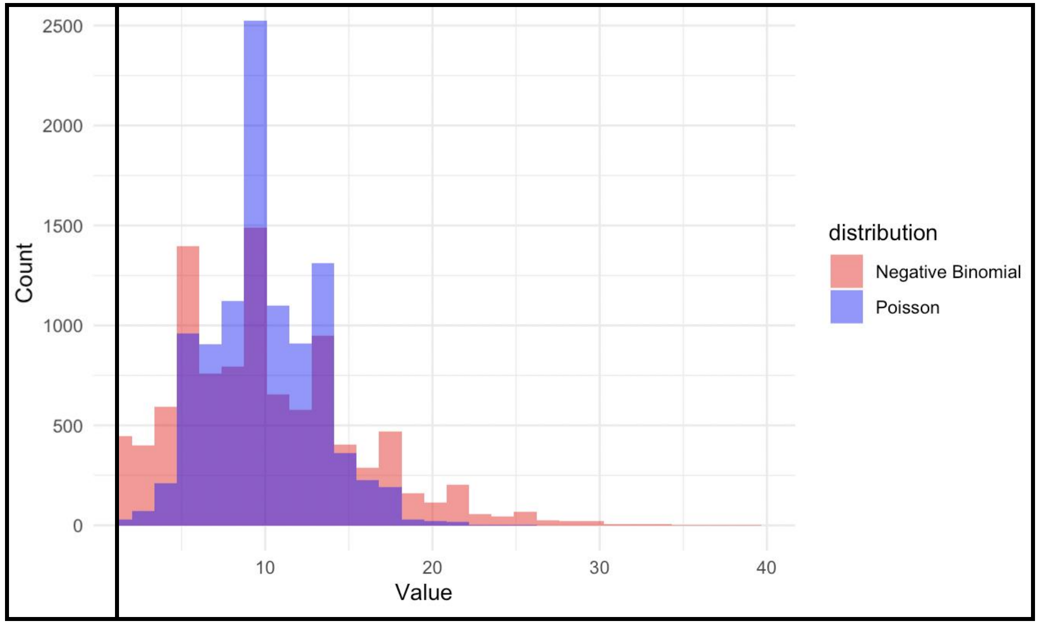

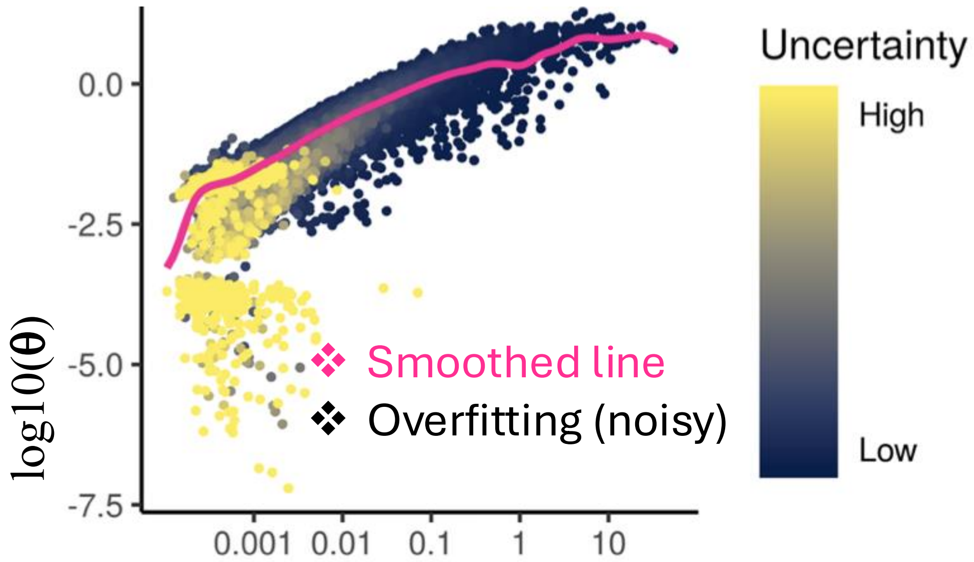

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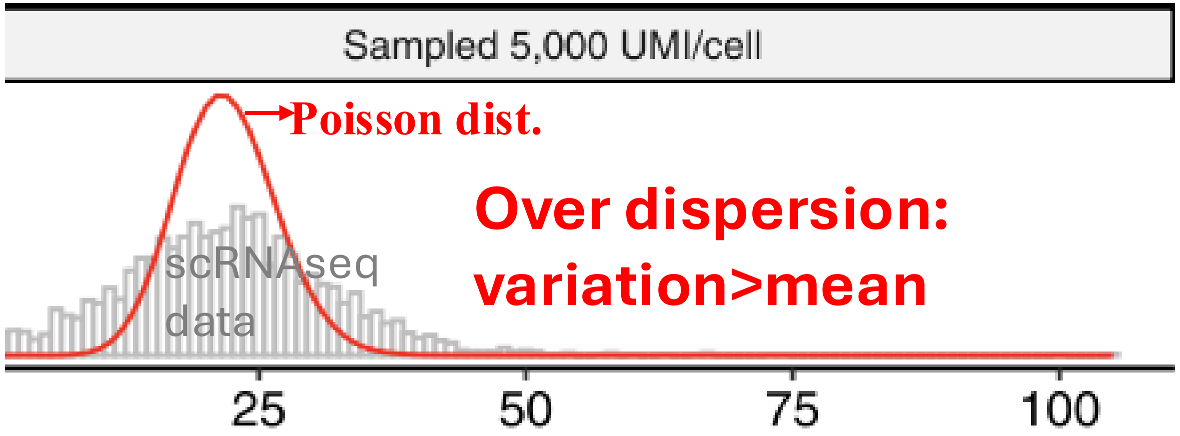

Over dispersion: variation>mean

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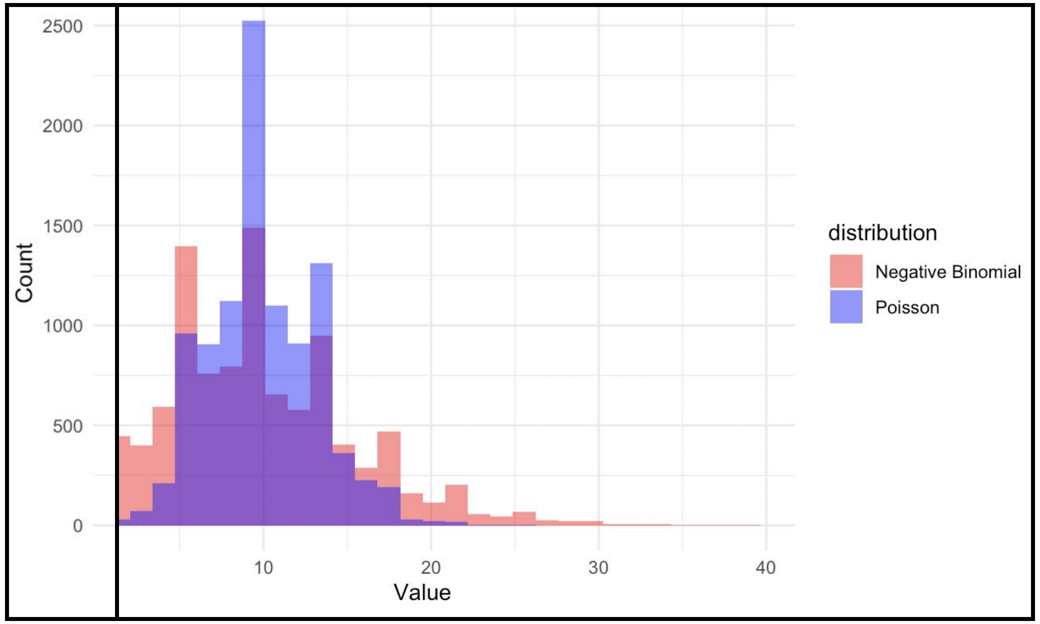
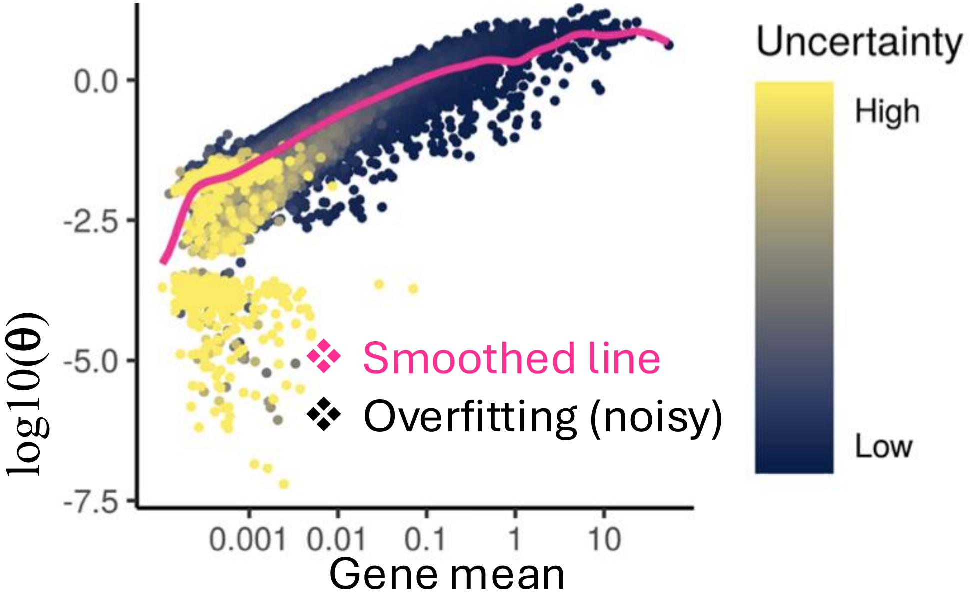


Model parameter Regularization

* Constrain towards smoothed line
* Increase estimate robustness

Exclude lowly expr. (Poisson) Genes (v2)



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$$\log(\mathbb{E}(x_i)) = \beta_0 + \boxed{\beta_1 \log_{10} m_i}$$

$$y_i = \mathrm{NB}(\mathrm{E}(x_i), \theta)$$

Dispersion parameter

Remove total cell UMI dependence

NB GLM:

Total cell UMI

$$\log(\mathbb{E}(x_i)) = \beta_0 + \boxed{\beta_1 \log_{10} m_i} + \text{Covariates}$$

Remove total cell UMI dependence

$y_i = \text{NB}(\text{E}(x_i), \theta)$

Dispersion parameter

y_{ij} observed count for gene i in cell j

μ_{ij} expected mean for gene i, cell j

β_{0i} intercept (baseline mean)

$\hat{\mu}_{ij} = \exp\left(\beta_0 + \beta_1 \log\left(\text{UMI}_j\right)\right)$

Pearson residual:

$$r_{ij} = \frac{y_{ij} - \hat{\mu}_{ij}}{\boxed{\sqrt{\hat{\mu}_{ij} + \hat{\mu}_{ij}^2/\theta_i}}}$$

NB SD

SCT:scale.data slot:

* No overdispersion
* Nearly normal
* mean = 0, variance stabilized
* total UMIs (+covariates) effect removed
* Ready for PCA

$$y_{ij}^{\text{corrected}} = r_{ij} * \text{NB SD} + \exp\left(\beta_0\right)$$

SCT:count slot:

* Neg. binomial distribution
* overdispersion (variance > mean)
* Variance increase with mean
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SCT:data slot:

* log transformed
* near normal
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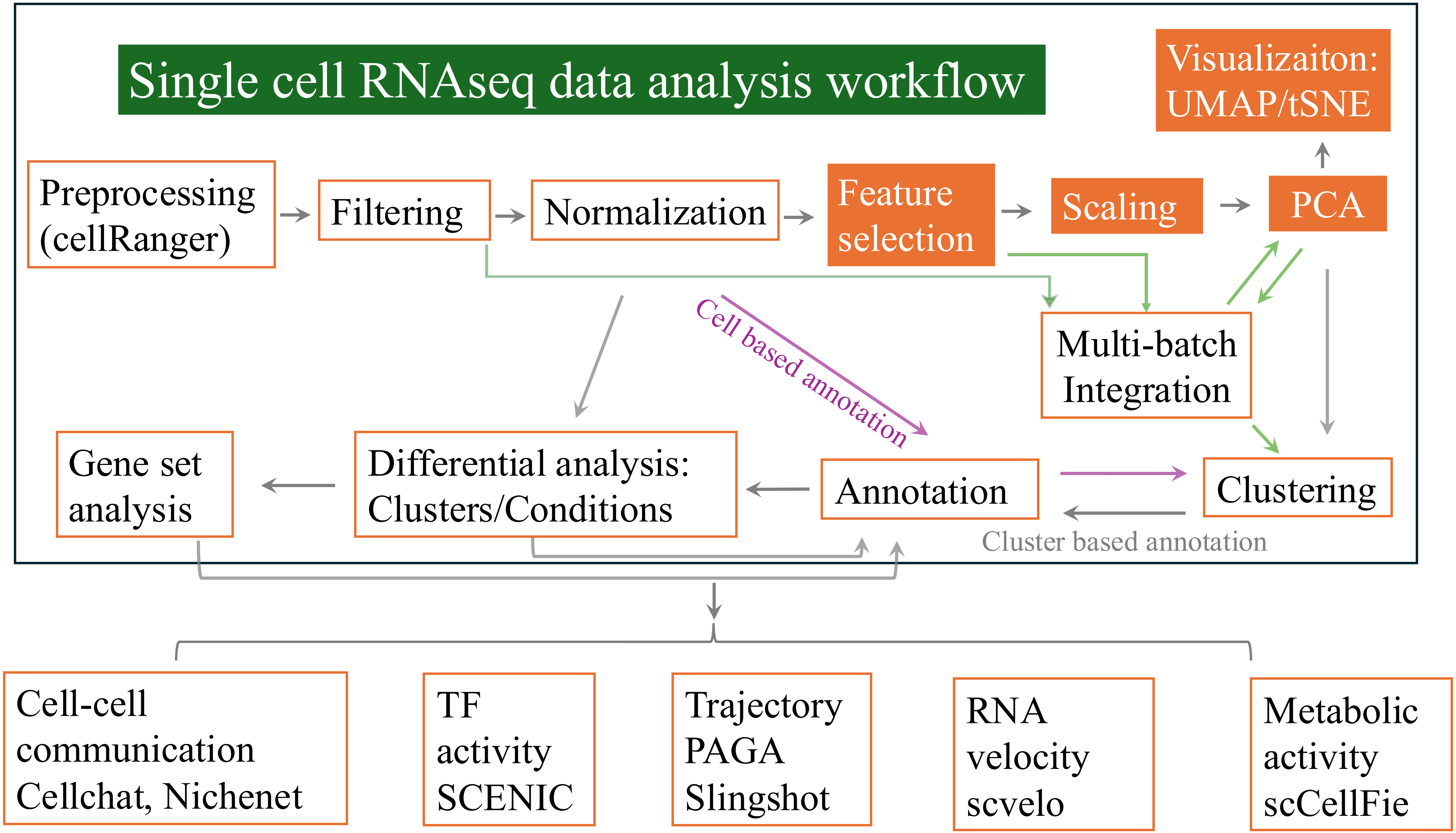
Dimensional reduction



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High dimensional curse

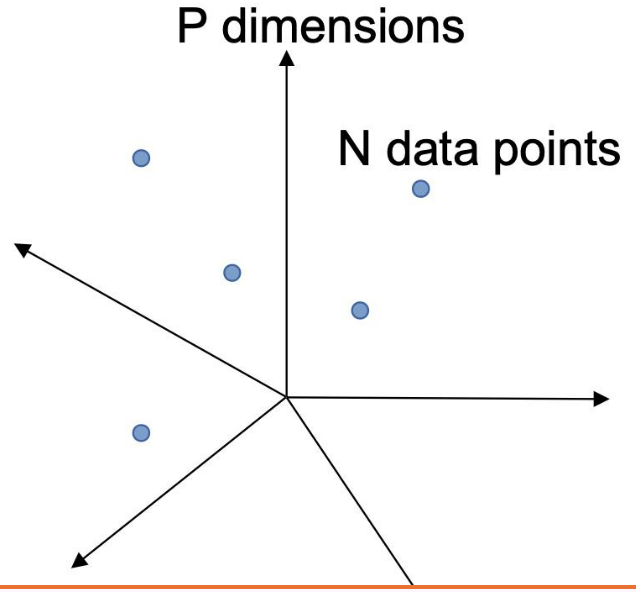
Single cell data: high dimensional data (P genes = P dimesions, thousands of genes)

Statistical observations: e.g. samples, cells etc. → N

Dimensions: P

Features: genes, proteins, microbes, metabolites etc.

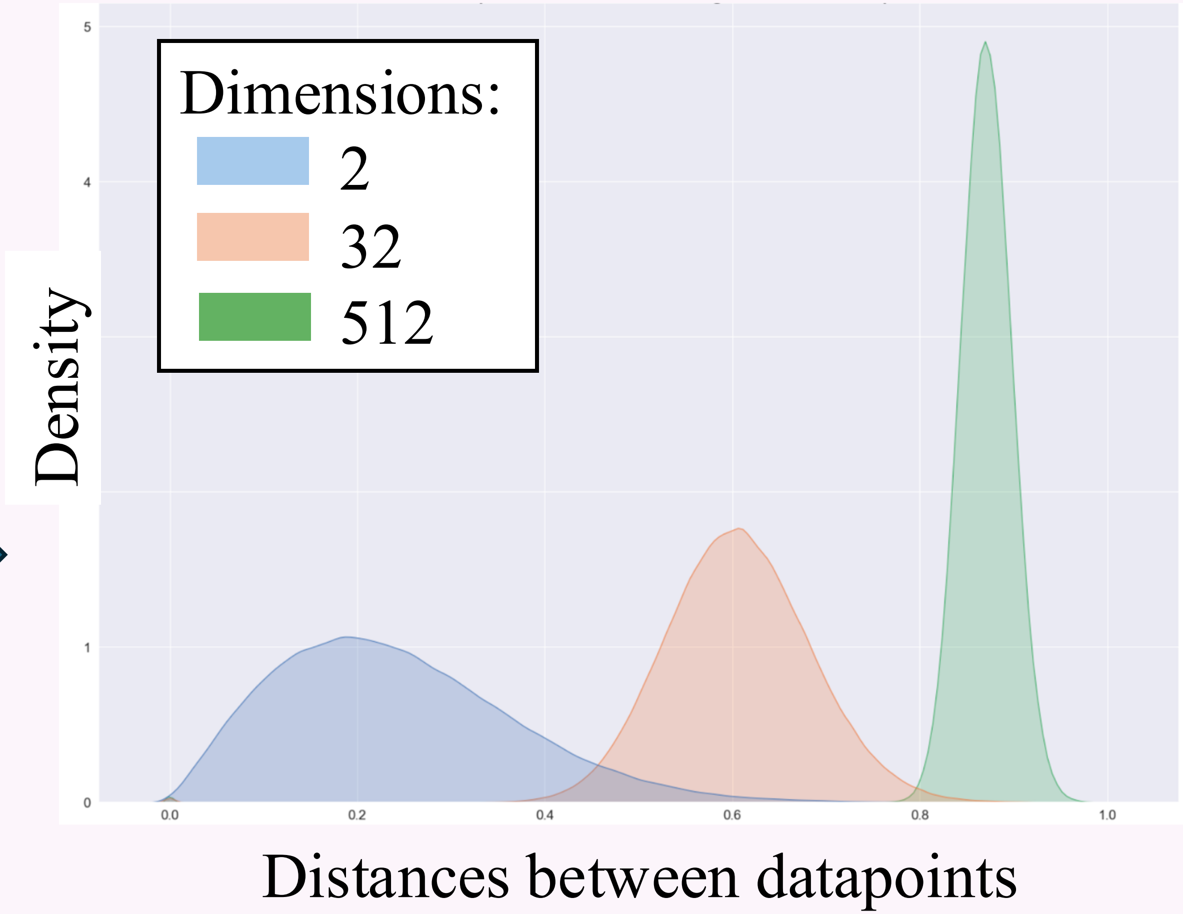
$$\begin{pmatrix} 0 & 3 & 1 & 0 & 2 & 3 & 8 & 1 & 1 & 3 \\ 1 & 1 & 0 & 0 & 7 & 1 & 2 & 2 & 3 & 3 \\ 1 & 2 & 2 & 0 & 0 & 6 & 7 & 1 & 2 & 2 \\ 1 & 2 & 3 & 10 & 0 & 4 & 6 & 1 & 0 & 5 \\ 3 & 2 & 2 & 1 & 4 & 3 & 2 & 1 & 6 & 0 \\ 7 & 4 & 4 & 5 & 3 & 9 & 6 & 1 & 6 & 1 \\ 7 & 1 & 1 & 5 & 2 & 8 & 9 & 1 & 3 & 6 \\ 5 & 0 & 1 & 6 & 2 & 0 & 0 & 0 & 1 & 5 \\ 1 & 6 & 3 & 3 & 4 & 6 & 2 & 0 & 1 & 1 \\ 1 & 2 & 2 & 4 & 1 & 1 & 3 & 0 & 8 & 2 \end{pmatrix}$$



High dimensional curse

Curses:

- Data sparsity (missing values, noisy)
- Multicollinearity (correlation between genes)
- Multiple testing
- Overfitting
- Data points very far from each other (**hard to form clusters/celltypes! Visualization**)



High dimensional curse

- **Solutions:**
  - Select (e.g. 2k) highly variable genes/HVGs (less genes/dimensions)
  - **Dimensional reduction**

    (**Compress** 2k HVGs into **fewer** dimensions/`MegaGenes`)

    * (Scaling)

    - PCA
    - tSNE
    - UMAP

Select highly variable genes

Assumptions:

- Features (i.e. genes) with low variance: **noises**
- Features with high variance: biological factors

- Seurat:sctransform()
  - Based on the residual variance
- Seurat:FindVariableFeatures()
  - Vst:
    - Model the mean and variance of the normalized data,
    - Select thos features deviating the most from the predicted variance
  - Dispersion:
    - Normalized by mean

Scaling

- Why do scaling?
  - PCA/tSNE/UMAP: multivariate analysis
  - Require features to be on similar scales

- Seurat SCT:scale.data
  - Pearson residuals (variance stabilized residuals)
- Seurat RNA:scale.data
  - With vars.to.regress
    - Standardized raw residuals
  - Without vars.to regress
    - Standardized normalized count (mean=0, variance=1)

RNA:count slot: raw count

Seurat::NormalizeData()

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RNA:data slot: normalized count

* Near normal distribution
* All cells the same total UMIs

Seurat::ScaleData()

With *vars.to.regress*

$= \mathbf{y} - \hat{\mathbf{y}}$

$$z_{ij} = \frac{x'_{ij} - \bar{x}_i}{s_i}$$

- Centered to mean=0
- Scale to variance=1

Without *vars.to.regress*

Covariates: e.g. mito

$$\text{Expression}_i = \beta_0 + \beta_1 \times \text{Var}_1 + \beta_2 \times \text{Var}_2 + \cdots + \varepsilon_i$$

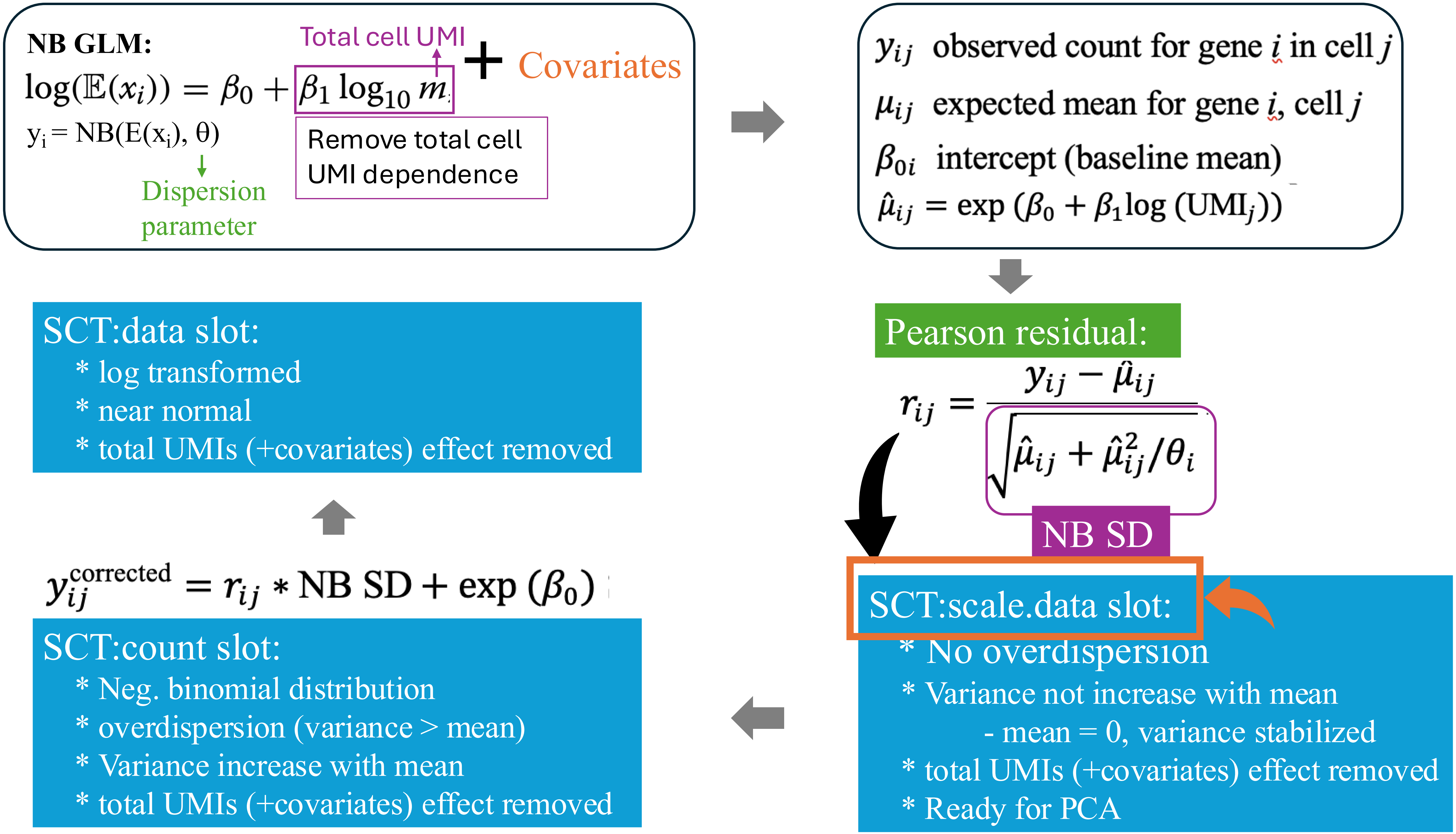
Get raw residual:

Standardize the residuals:

* center: mean=0
* scale: variance=1

RNA:scale.data slot:

* mean = 0, variance=1
* total UMIs (+covariates) effect removed



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SCT:data slot:
* log transformed
* near normal
* total UMIs (+covariates) effect removed

Dimensional reduction

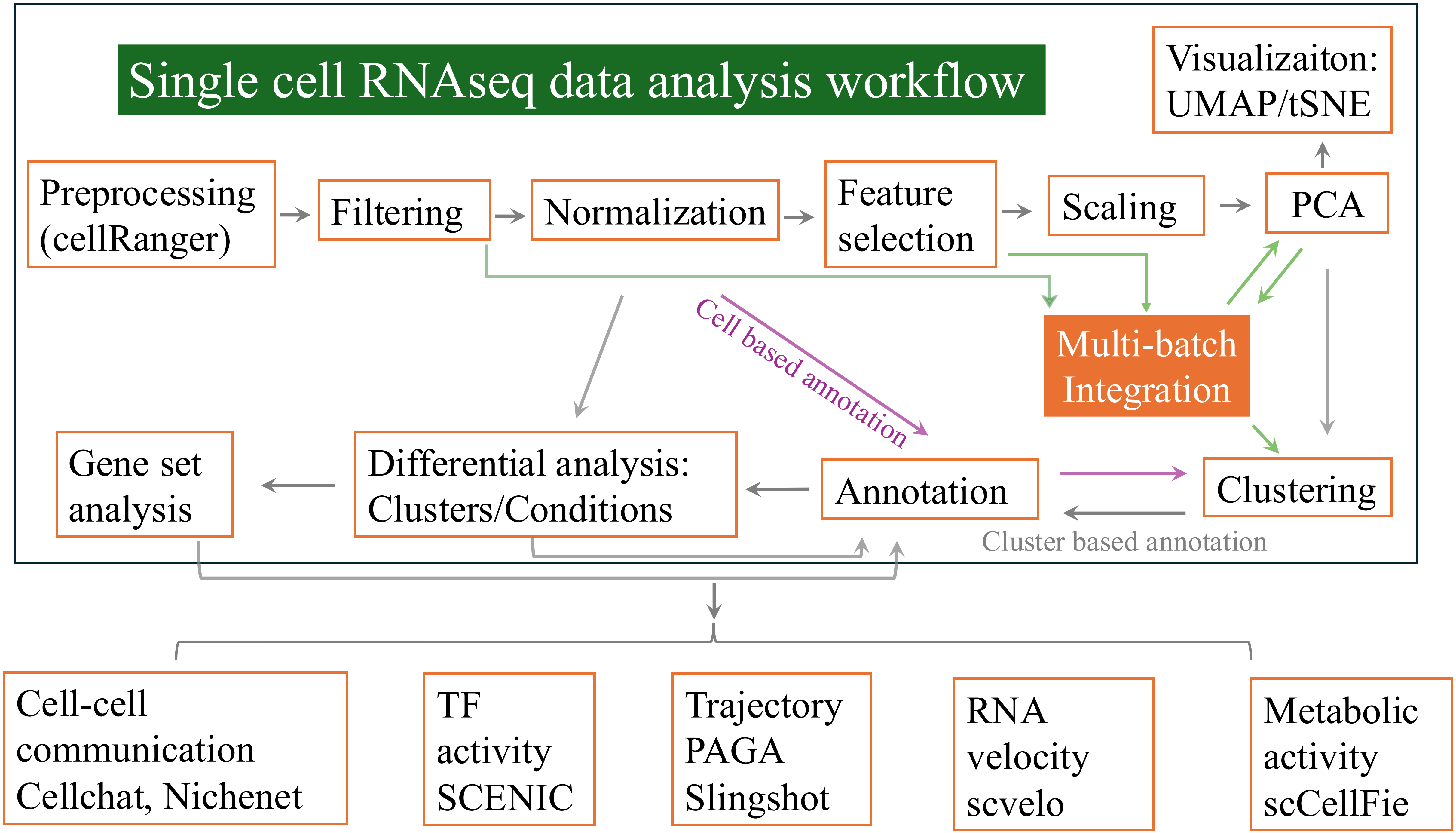
- **PCA**:
  - Linear transf.(accurately represent distances between cells)
  - Cannot efficiently pack differences in high dim. into the first 2 PCs (principal components)
- **tSNE**
  - Preserve local structure
  - Perplexity, low: finer structue
  - Slow
- **UMAP**:
  - Preserve both local & global structure
  - n_neighbors, min_dist, low: finer structure
  - Fast
- **tSNE&UMAP**
  - Non-linear (discard inf. distort distance)
  - NOT accurately represent distances between cells

Quantification

Only for visualization

Moon et al. 2019 Nature Biotechnology

What if there are multiple sampels from different batches?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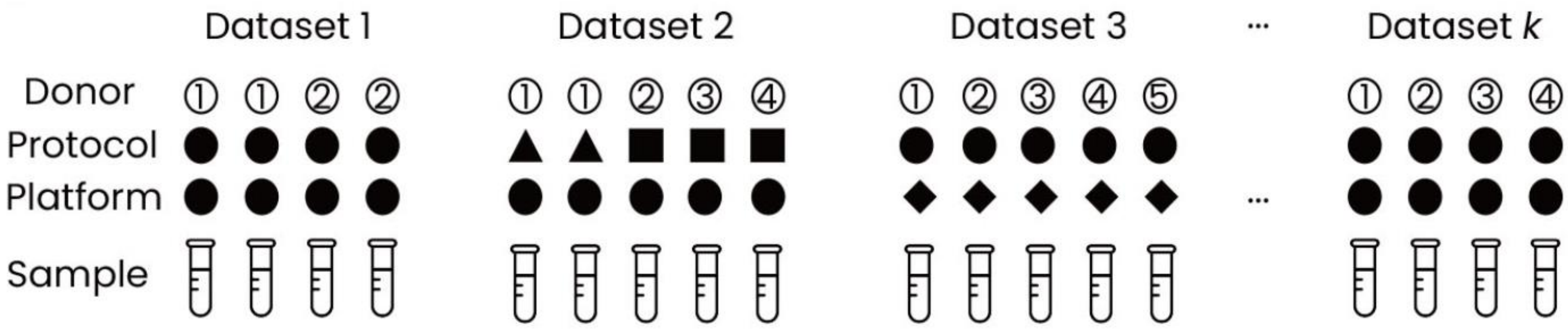
Integration: remove batch effect

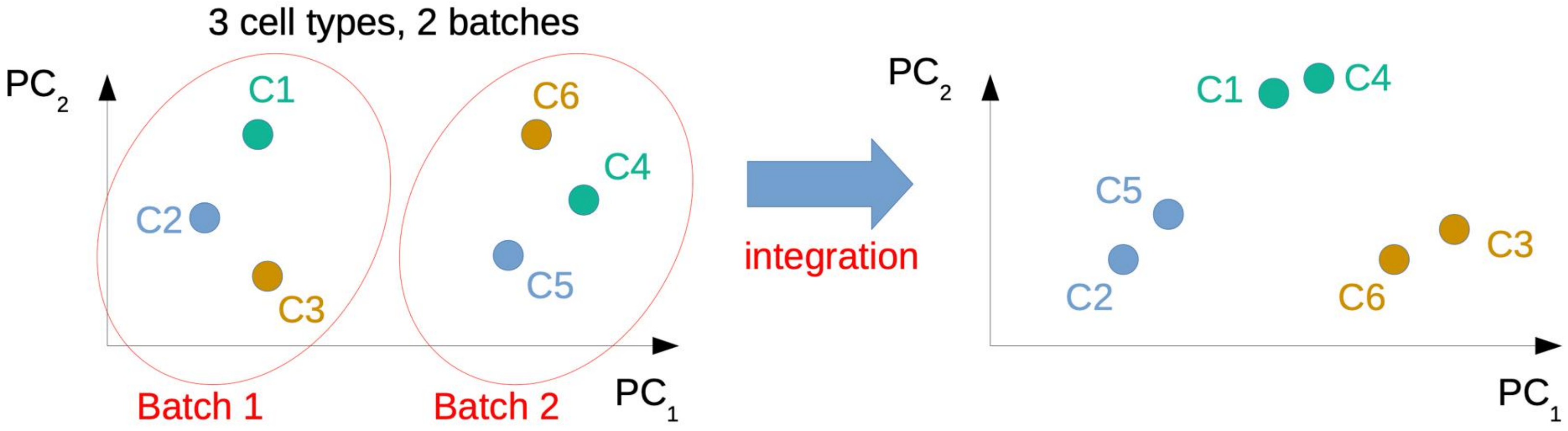
Batches:

- Sample characteristics (donors, tissues, species, disease conditions) …
- Experimental protocols, laboratories, sample acquisition and handling, sample composition, reagents or media and/or sampling time …
- Sequencing platforms/lanes/flow cells …



Ryu et al. Mol. Cells 2023; 46(2): 106-119

Integration: remove batch effect

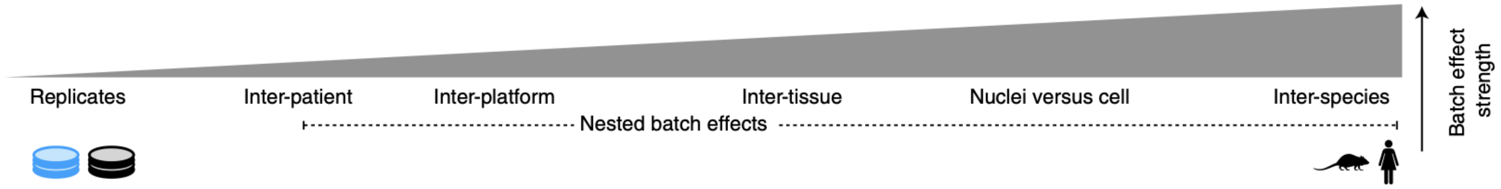


a

	Considerations	scANVI	Scanorama embed	scVI	FastMNN embed	scGen	Harmony	FastMNN gene	Seurat v3 RPCA	BBKNN	Scanorama gene	ComBat	MNN	Seurat v3 CCA	trVAE	Conos	DESC	LIGER	SAUCIE embed	SAUCIE gene
Input	Programming language	Python	Python	Python	R	Python	R	R	R	Python	Python	Python / R	Python / R	R	Python	R	Python	R	Python	Python
	Method runs without additional information	✗				✗														
Scib results	Consistent top performer	✓	✓	✓		✓														
	Top method on small/ simple tasks		✓		✓	✓	✓													
	Top method on large/ complex tasks	✓	✓	✓		✓														
	Top method on ATAC data	–		–			✓											✓		
Task details	Integrates strong batch effects	✓	–	–		✓			–	–				–						
	Top method for recovery cell states or modules	✓	✓								✓	✓	✓							
	Confounding of bio and batch variance	✓	–			✓														
	Top method for trajectories	–	✓	–	✓	✓														
	Method deals with varying compositions											✗								
Speed	Fast method for quick results									✓		✓								
	Scales well to large datasets on CPU	✓	–	✓						✓	–								✓	✓
	Method has GPU support	✓		✓		✓									✓		✓		✓	✓
	Scales well to feature spaces beyond genes														✓	✓				
Output	Method shows corrected expression					✓		✓	✓		✓	✓	✓	✓						✓
	Method gives relative cell embeddings									✗						✗				

✓ Fulfills the criterion
– Partial fulfillment of criterion
✗ Does not fulfill criterion
Python
R

b



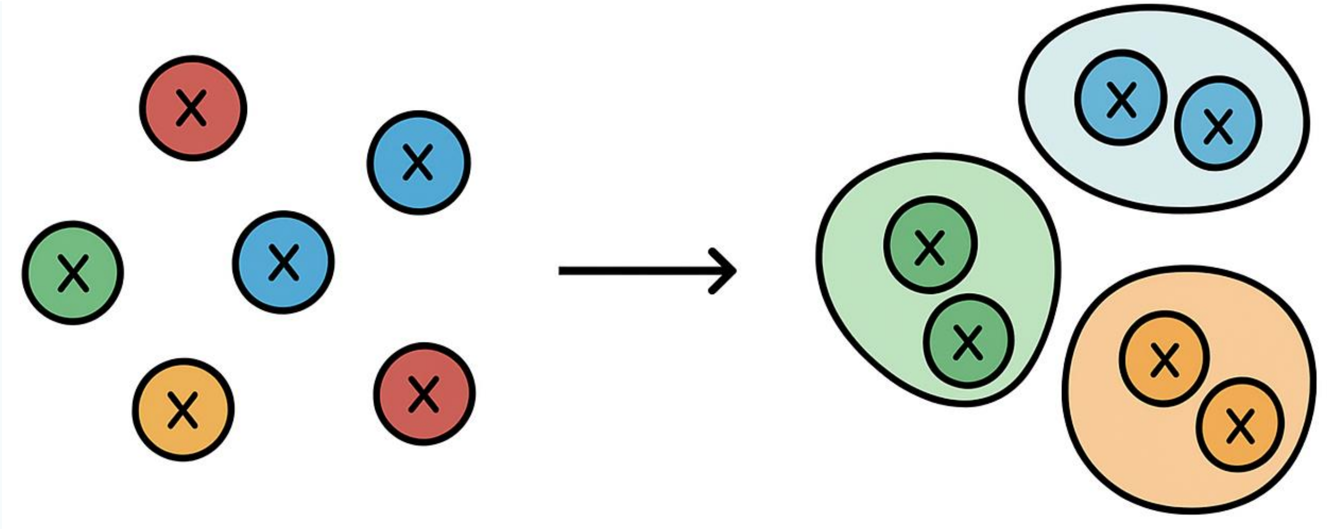
Lueken et al. 2022 Nature Methods

Graph-based

Clustering

Analyze ind. cells?

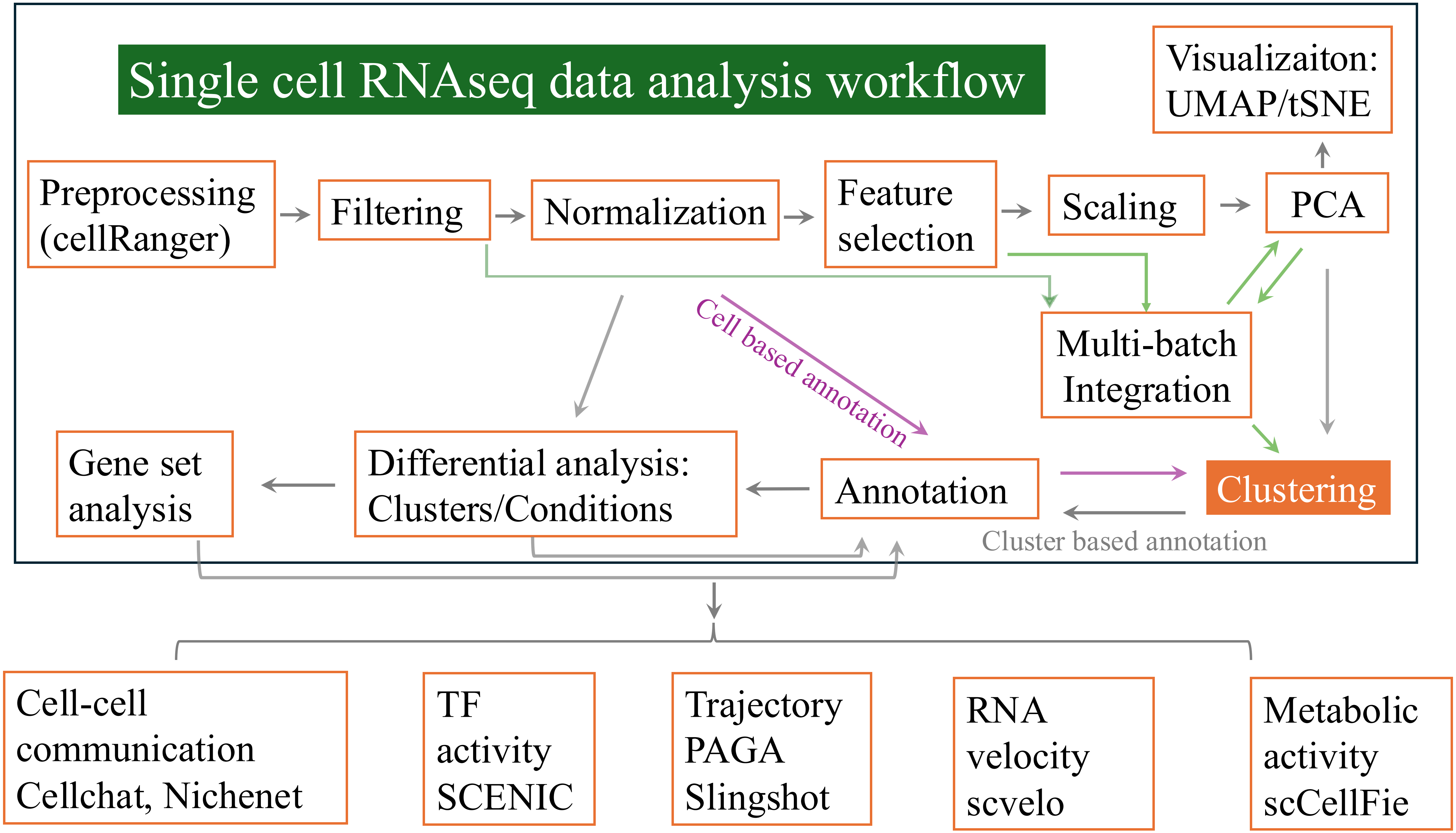
Group cells to clusters!



- Noise,variability
- Biological context lost

- Reduce noise, robust insights
- Biologically meaningful patterns

Cell clusters/groups $\approx$ Celltypes/cellular states



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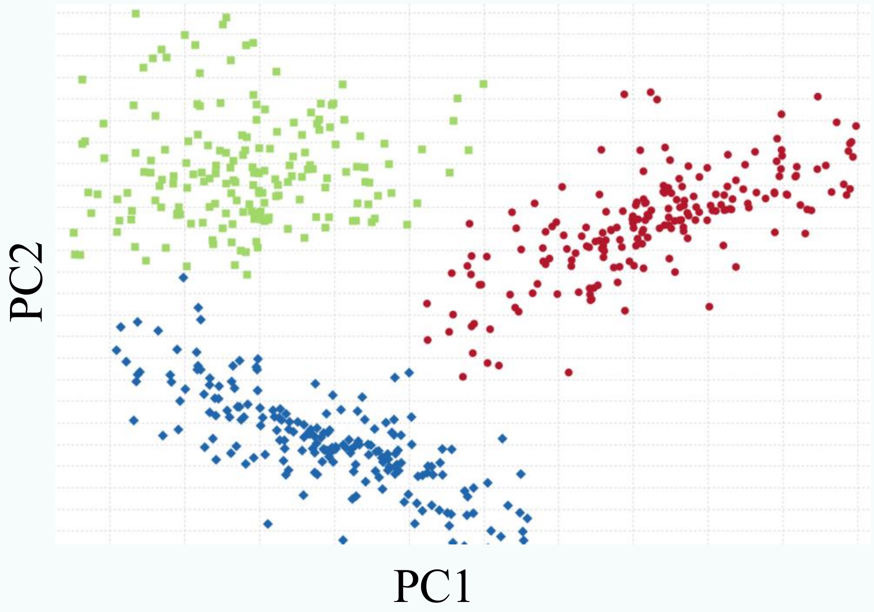
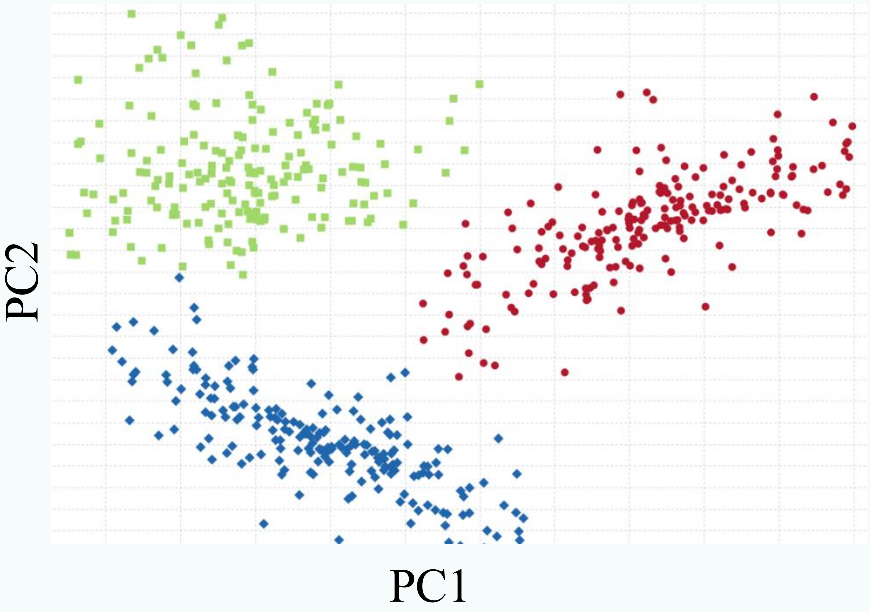
PCA space: on top n PCs

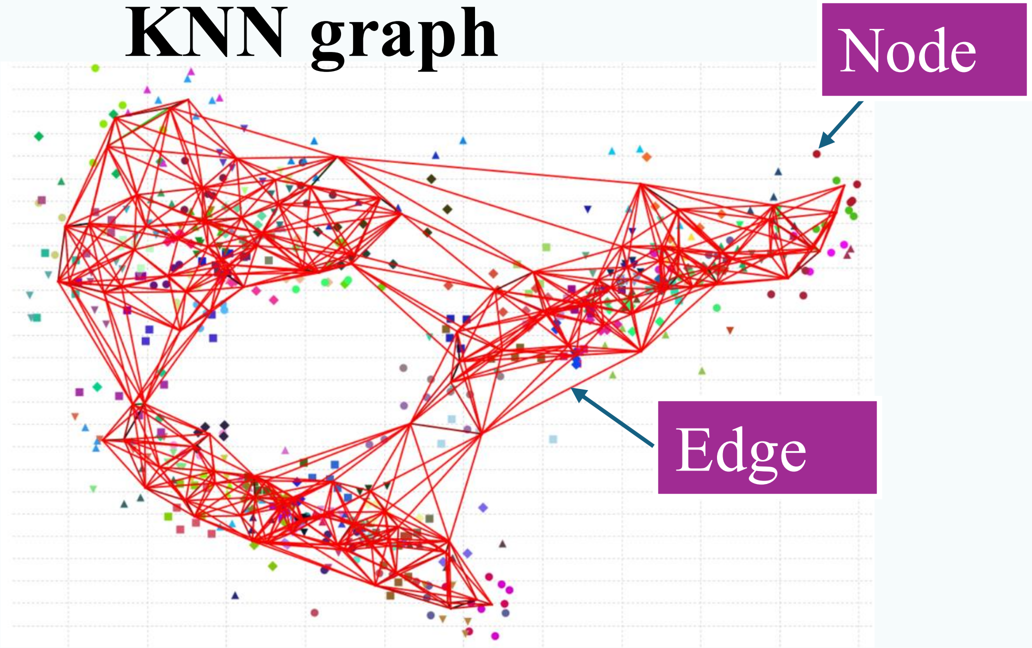


PCA space: on top n PCs



Connect K nearest neighbors

KNN graph



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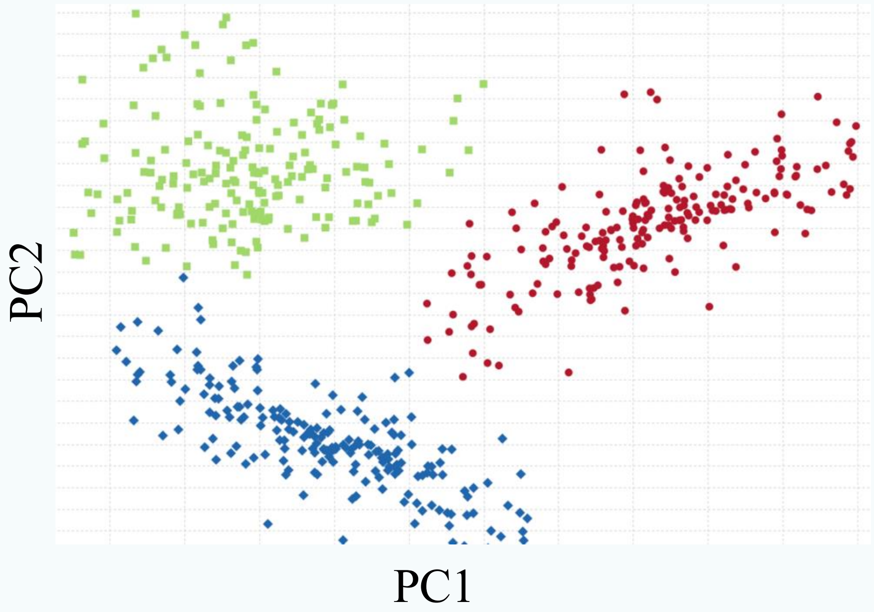

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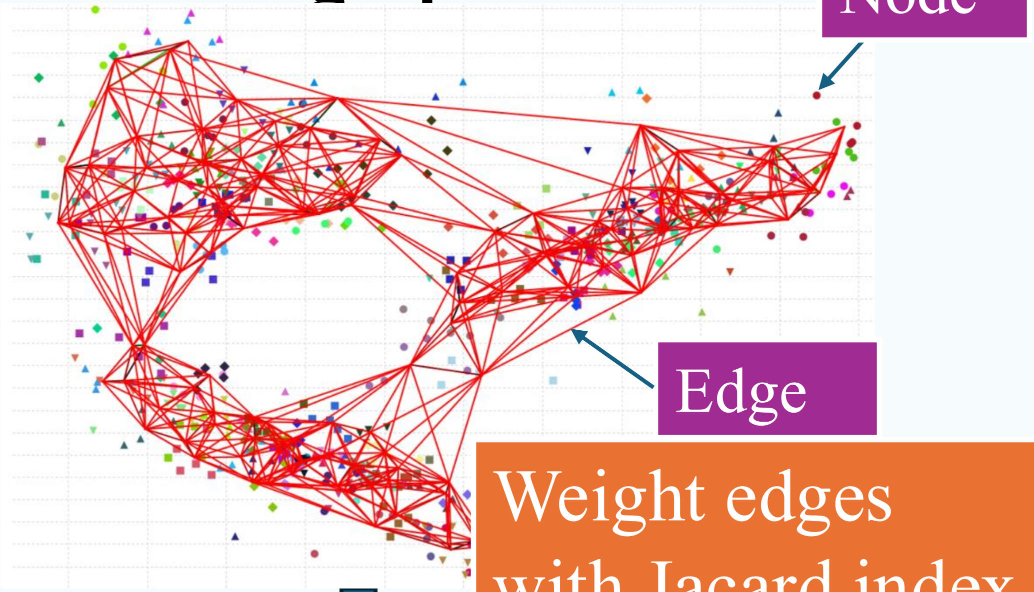
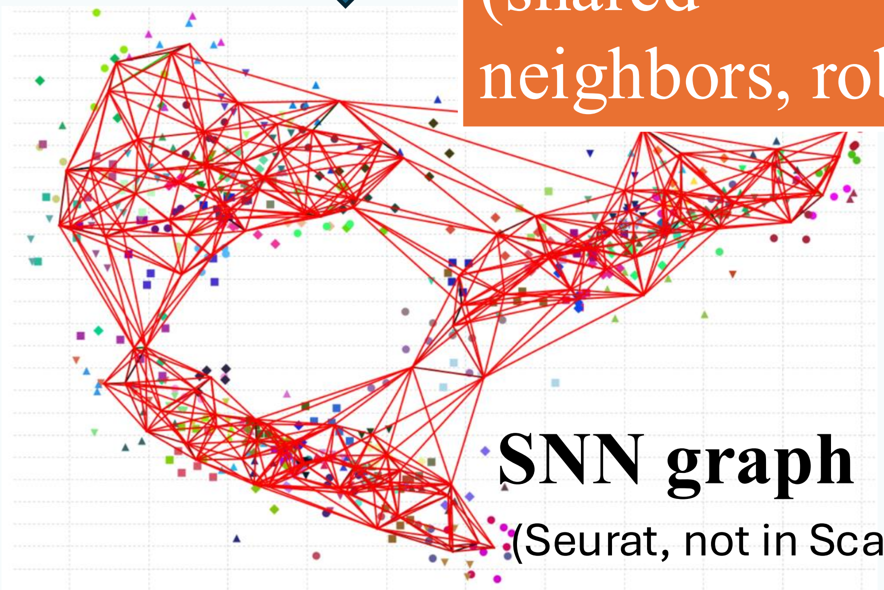


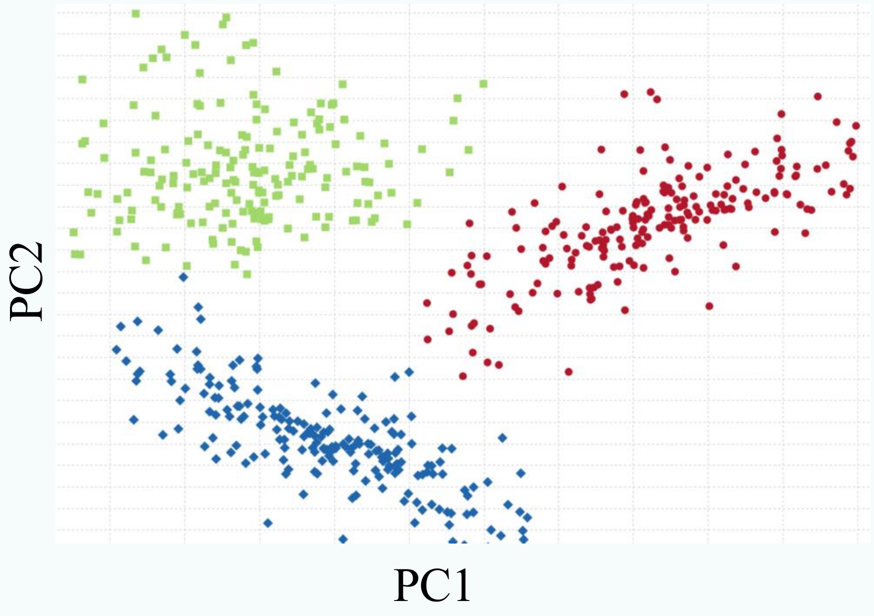
Weight edges with Jacard index (shared neighbors, robust)



SNN graph

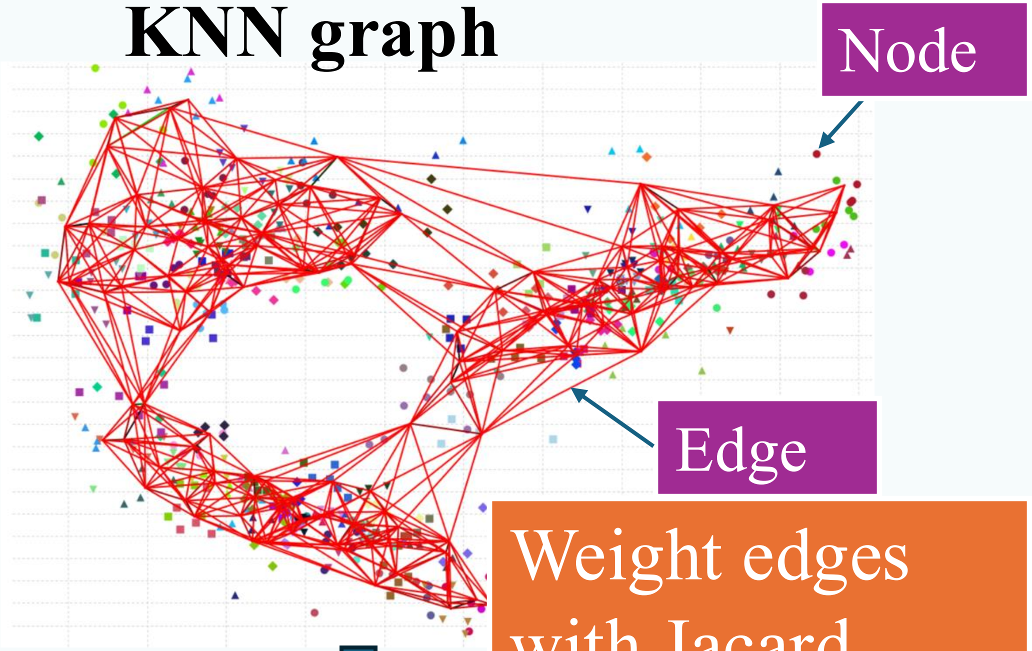
(Seurat, not in Scanpy)

PCA space: on top n PCs

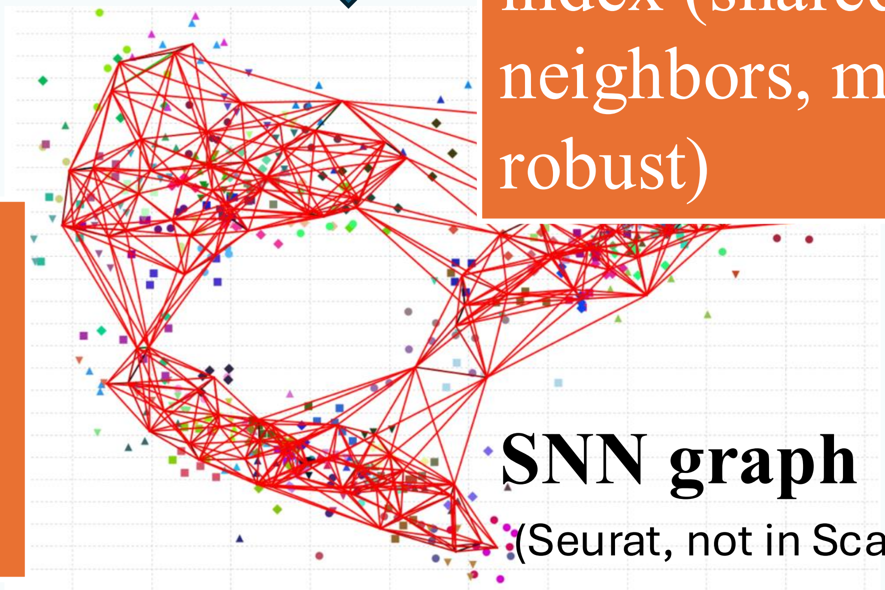


Connect K nearest neighbors

KNN graph



Weight edges with Jacard index (shared neighbors, more robust)

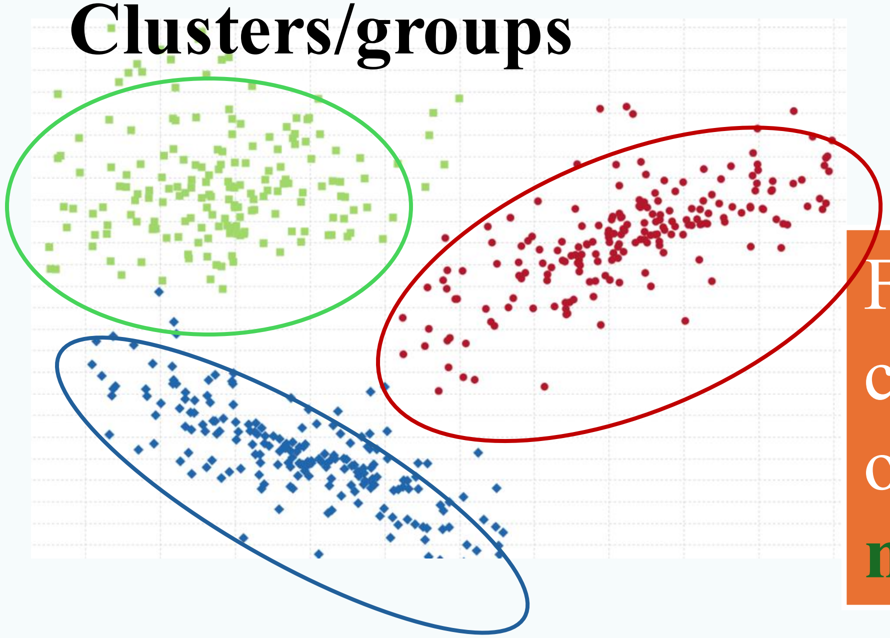


SNN graph

(Seurat, not in Scanpy)

Find cell communities by optimizing **modularity**

Clusters/groups



Modularity

Maximize the difference between the observed edges within clusters and the expected edges

$$\mathcal{H} = \frac{1}{2m} \sum_c \Big(\underbrace{e_c}_{\text{Observed edges within cluster c}} - \gamma \underbrace{\frac{K_c^2}{2m}}_{\text{Expected edges within cluster c}} \Big)$$

Resolution: higher, more clusters

Tragg et al. 2019 Scientific reports

Modularity

Maximize the difference between the observed edges within clusters and the expected edges

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Resolution: higher, more clusters

Observed edges within cluster c

Expected edges within cluster c

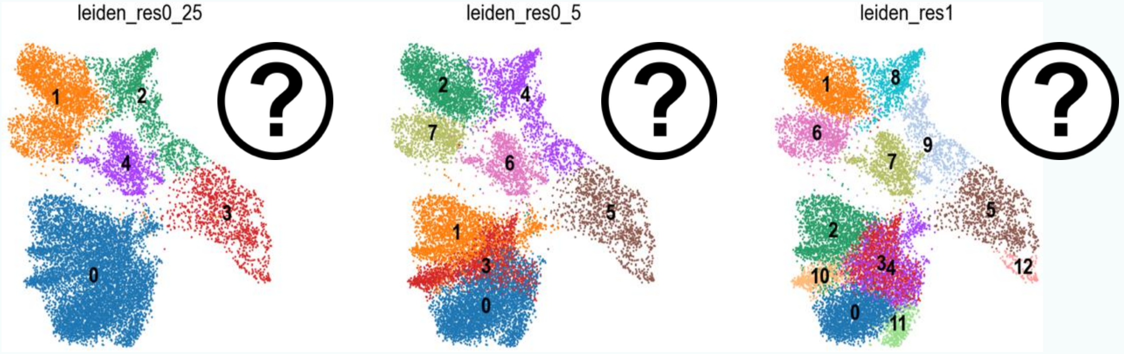
Modularity optimization algorithms:

Louvain
Leiden

Leiden: an improved version of Louvain

Tragg et al. 2019 Scientific reports

Clustering resolution, quality evaluation

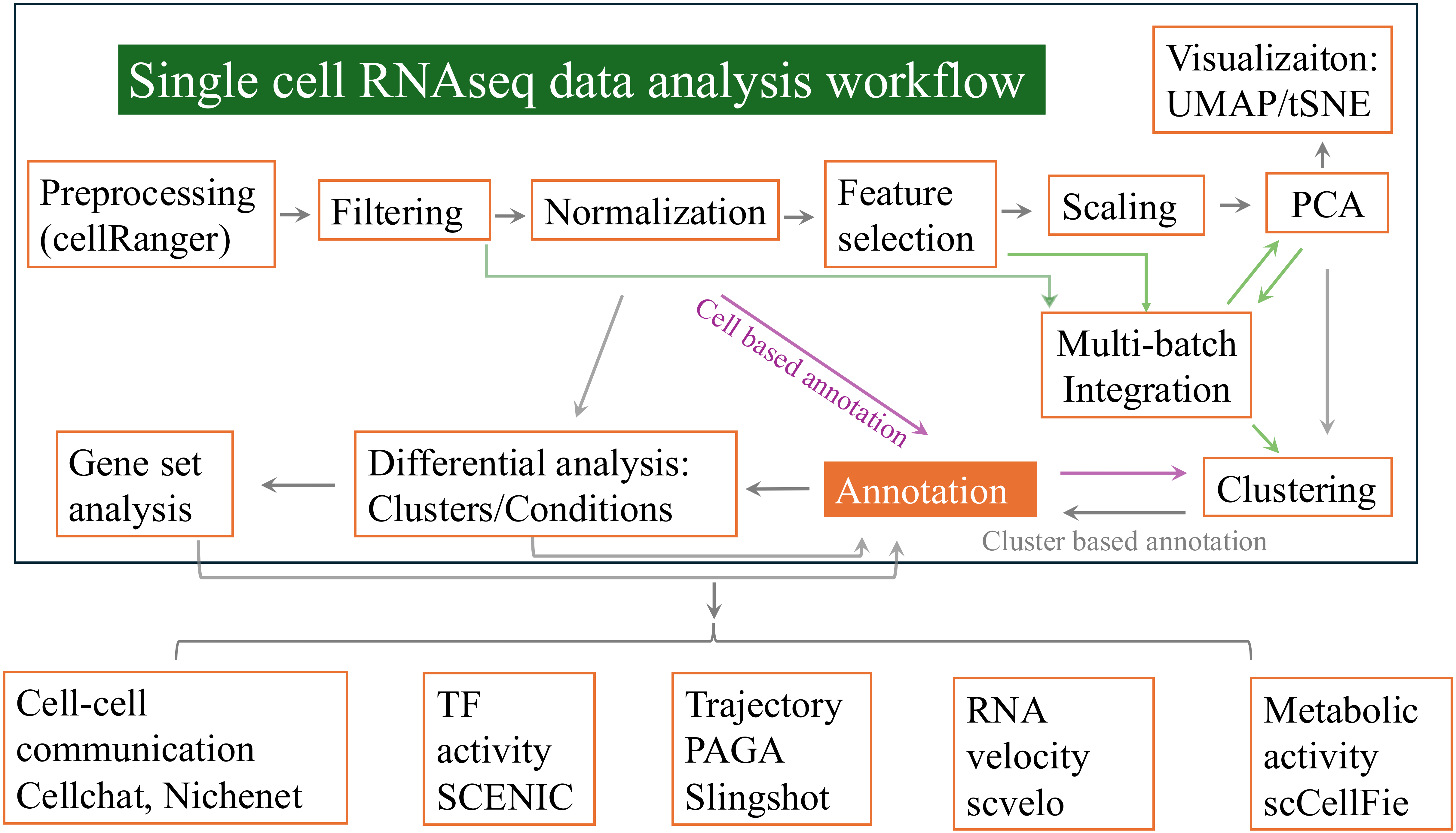


- **Resolution too low**: rare cell types merge with larger clusters, subtypes merge together
- **Resolution too high**: one celltype split to multiple clusters

- Measures
  - **Silhouette scores**
  - Calinksi-Harabasz index
  - Normalized mutual information
  - Variation of information
  - Graph-sensitive indices

- Cell-based automatic annotation before clustering
- The presence of cluster-specific differentially expressed genes
- Biology

https://support.sas.com/resources/papers/proceedings19/3409-2019.pdf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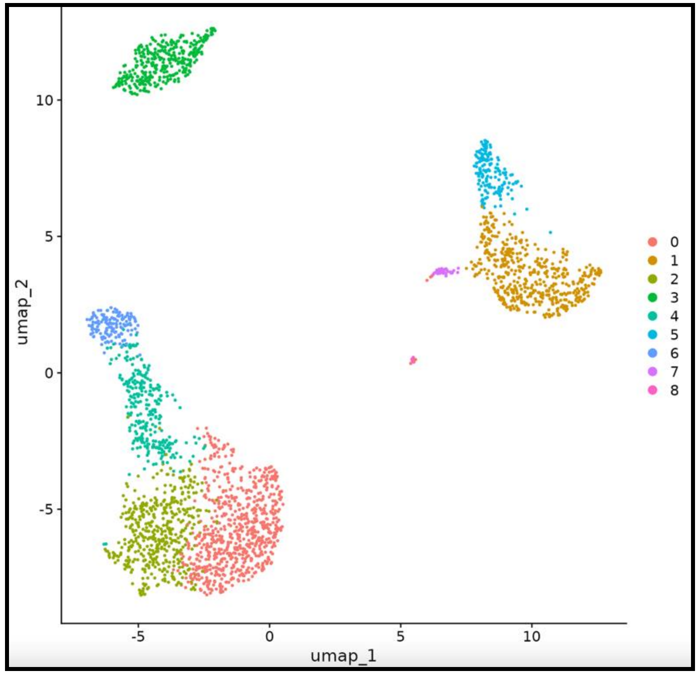
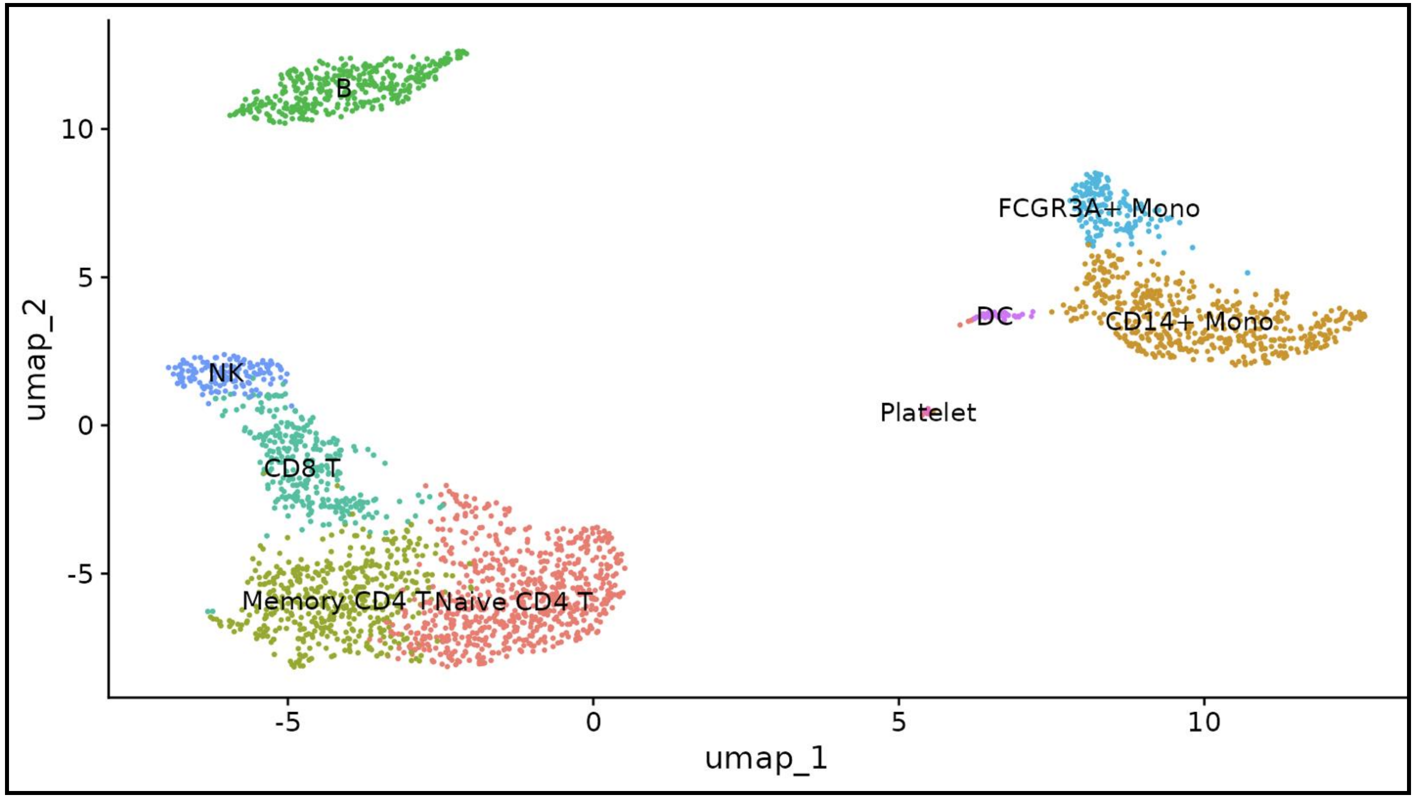
Cell annotation

Cell annotation: the process of labeling cells in your data with celltypes





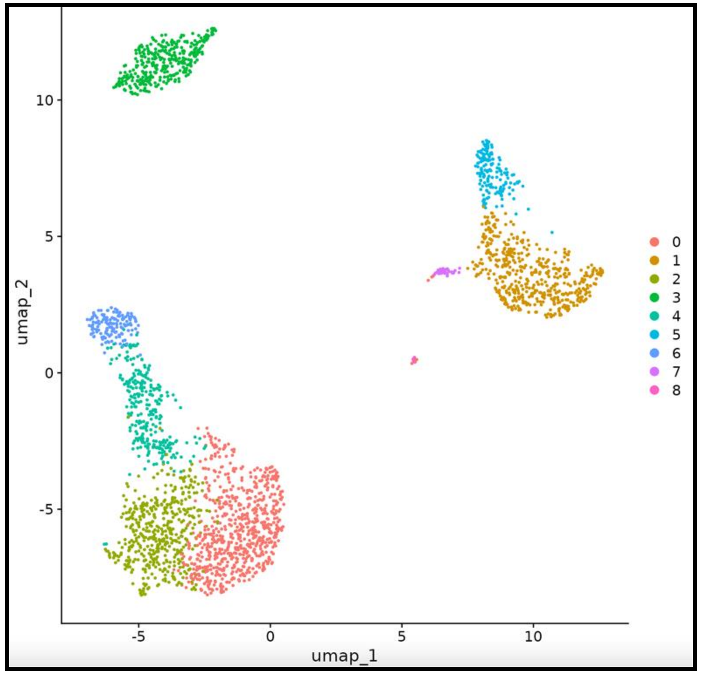
Celltype

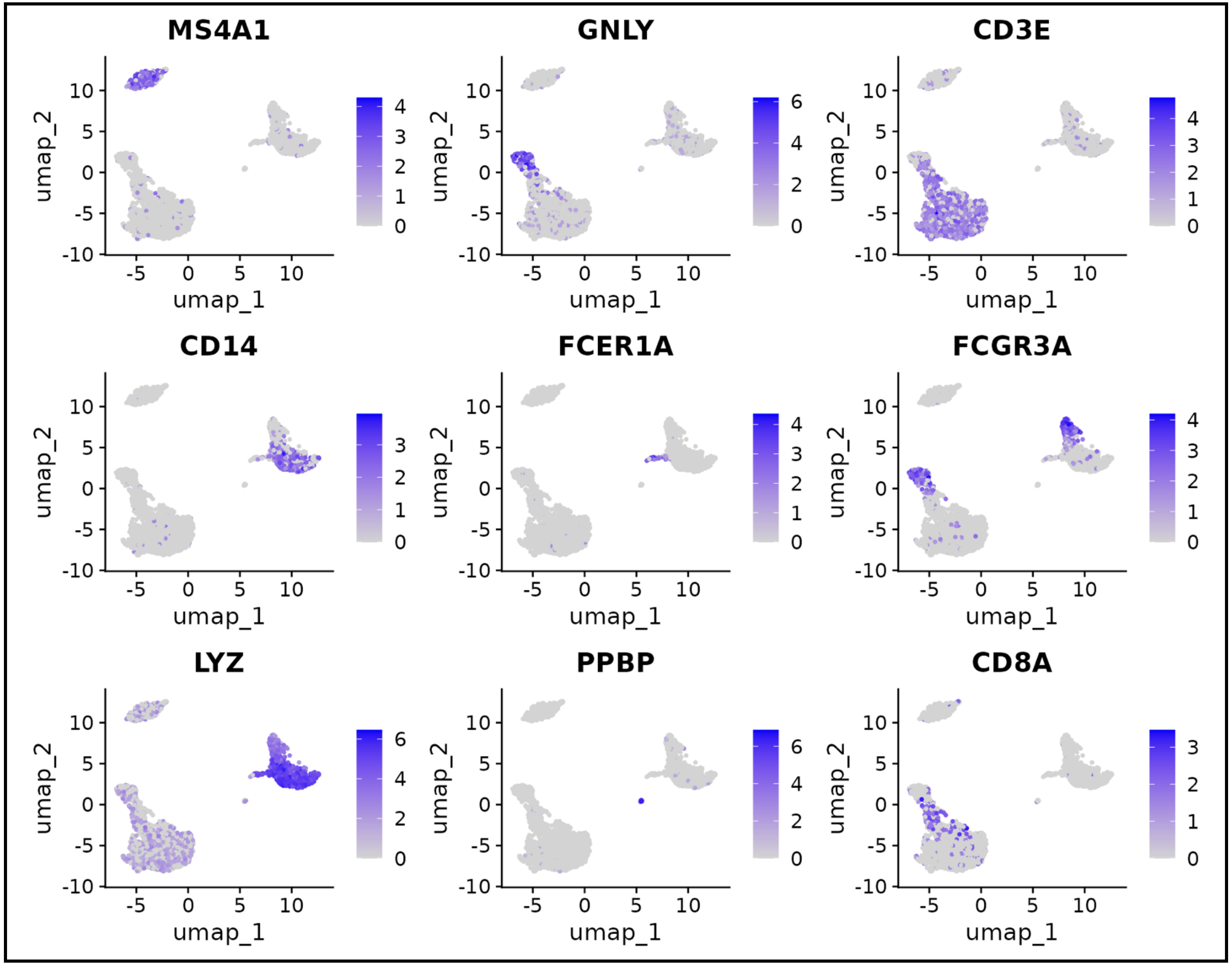
- A cellular phenotype that is robust across datasets
- Identifiable based on expression of specific markers (i.e. proteins or gene transcripts)
- Often linked to specific functions
- E.g. a plasma B cell: a type of white blood cell that secretes antibodies used to fight pathogens and it can be identified using specific markers

Manual annotation

- Manually annotate cells based on a single or small set of marker genes (*better from RNAseq paper, not from protein study*) known to be associated with a particular cell type.
- Transparent, novel celltypes
- **Two angles:**
  - 1. Start from a table of marker genes for all expected celltypes, check their expression in your dataset
  - 2. Check the highly expressed genes/specifically expressed genes in your clusters, see which ones are associated with known celltypes

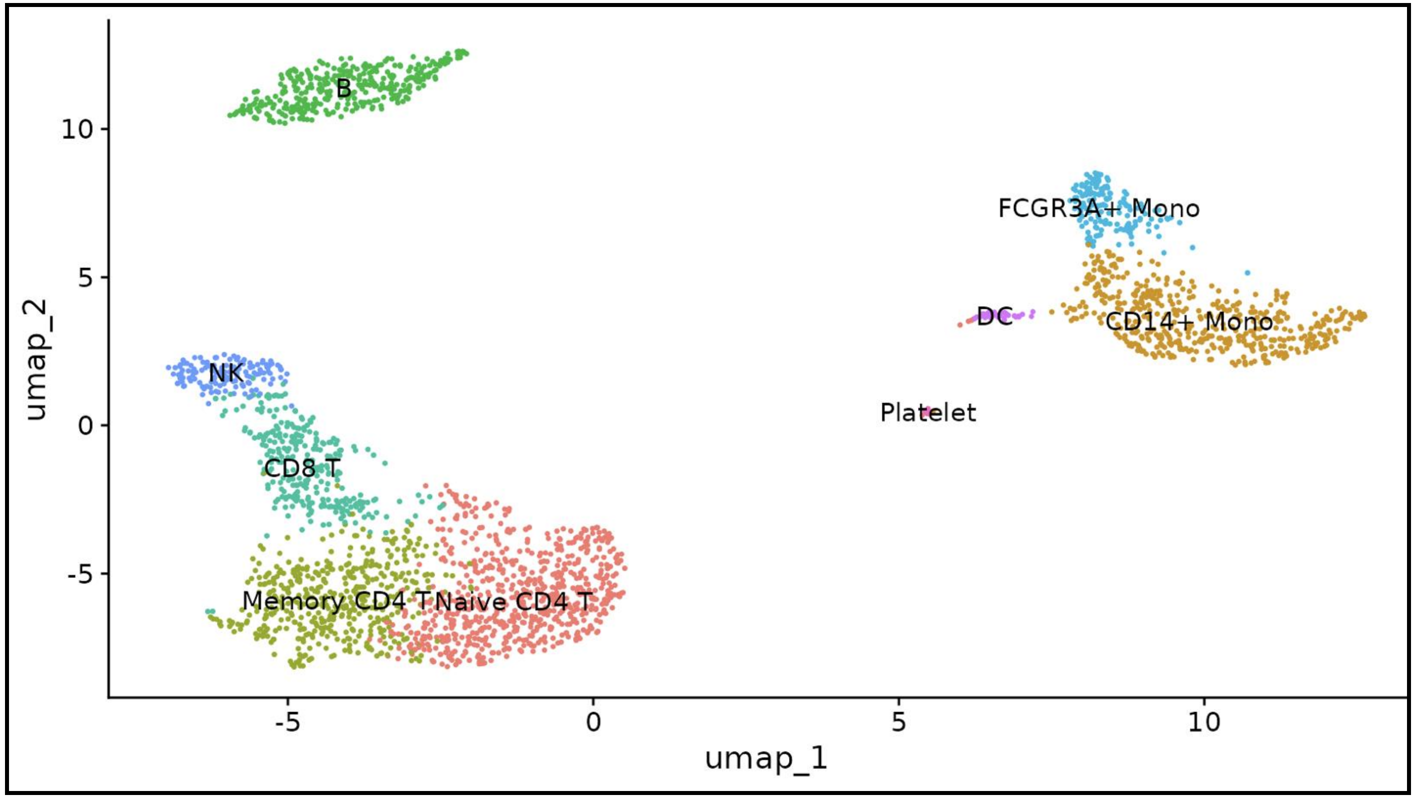
Manual annotation





- ➢ 1. Cluster cells into groups
- ➢ 2. Check gene expr.

Manual annotation



- 3. Assign celltype labels to clusters

Clarke et a. 2021 Nature

Manual annotation

Potential Caveats	Recommendation
Slow, labour-intensive.	**Whenever possible, begin with automatic annotation to determine general cell labels.**
Subjective.	**Work with an expert; consider multiple cell-type conclusions.**
Cell types not distinguishable by a single marker.	Use multiple markers for each cell type.
Known markers not distinguishing cell types.	Curate larger lists of markers from the literature, additional experiments or experts.
Conflicting marker gene sets between sources.	Select a marker gene set that best represents the biological signal being looked for in the data (e.g. if looking for cell subtypes, use more extensive gene sets than what is used for general cell-type annotation).

Clarke et a. 2021 Nature

Automatic annotation

- Fast
- Reproducible

Automatic annotation: computer algorithm + prior biological knowledge

- Quickly identify known celltypes
- Highlight unknown celltypes for further investigation (new, subtypes)

⬇

Cell annotation

- Ideal (will not miss diff. among cells)
- Low seq. depth: zeros (expressed, not detected), sparse, not enough information single cells

Cell cluster (group) annotation

- More accurate, robust
- Clustering quality
- Compare multiple methods

Clarke et a. 2021 Nature protocols

Automatic annotation: computer algorithm + prior biological knowledge

- Quickly identify known celltypes
- Highlight unknown celltypes for further investigation (new, subtypes)

Marker based

- **Known marker genes sources**: SCSig, PanglaoDB, CellMarker, DISCO, literature
- Conflict between sources
- Tools: AUCell, SCINA, GSEA/GSVA

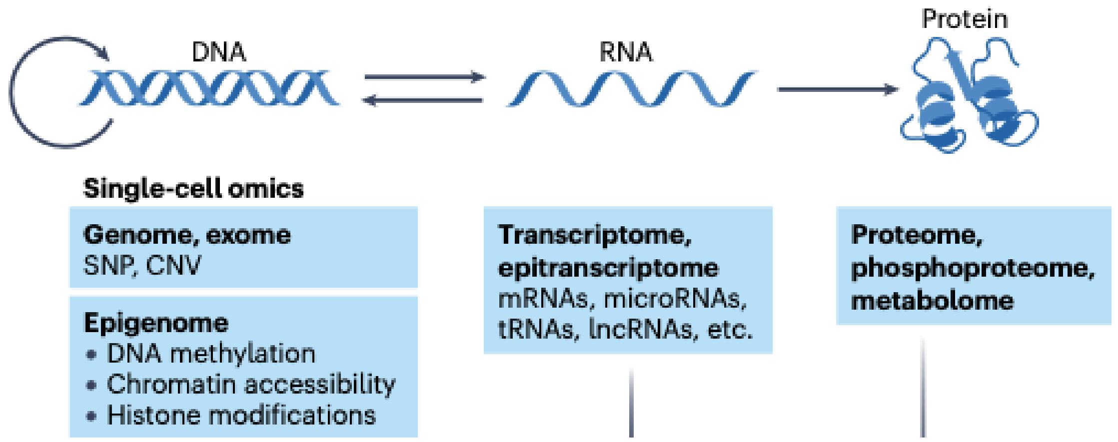
Reference dataset based

- **Query data**: your data, to be annotated
- **Reference data**: similar, expert-annotated dataset
- **Reference sources**: GEO, Single Cell Expression Atlas, cell atlas projects, DISCO, CellTypist, Azimuth, literature, Gut (https://www.gutcellatlas.org/pangi.html), cell ontology (https://cell-ontology.github.io/)
- Tools: Scmap, singleCell Net, singleR, Seurat

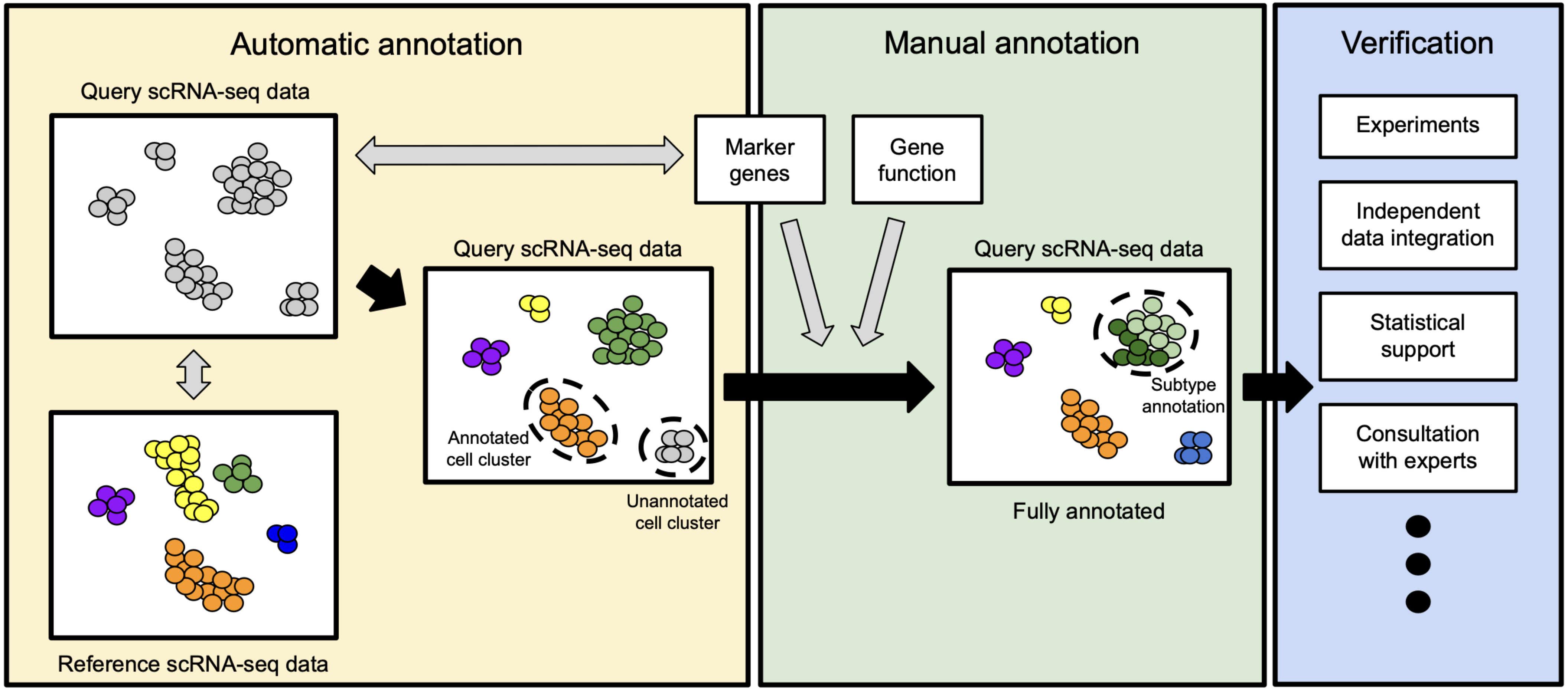
Clarke et a. 2021 Nature protocols

Annotation verification

- Statistical methods (e.g. Posterior inclusion probabilities, Chung 2020)
- Experimental validation:
  - Funcitonal assays (e.g. cytokine secretion, proliferative capacity),
  - Fluoresence in situ hybridization of source tissue sampels,
  - Validate marker gene expression in a larger number of samples
- Complementary genomic methods
  - Proteome, metabolome
  - Genome (SNP)
  - Epigenome
  - Spatial

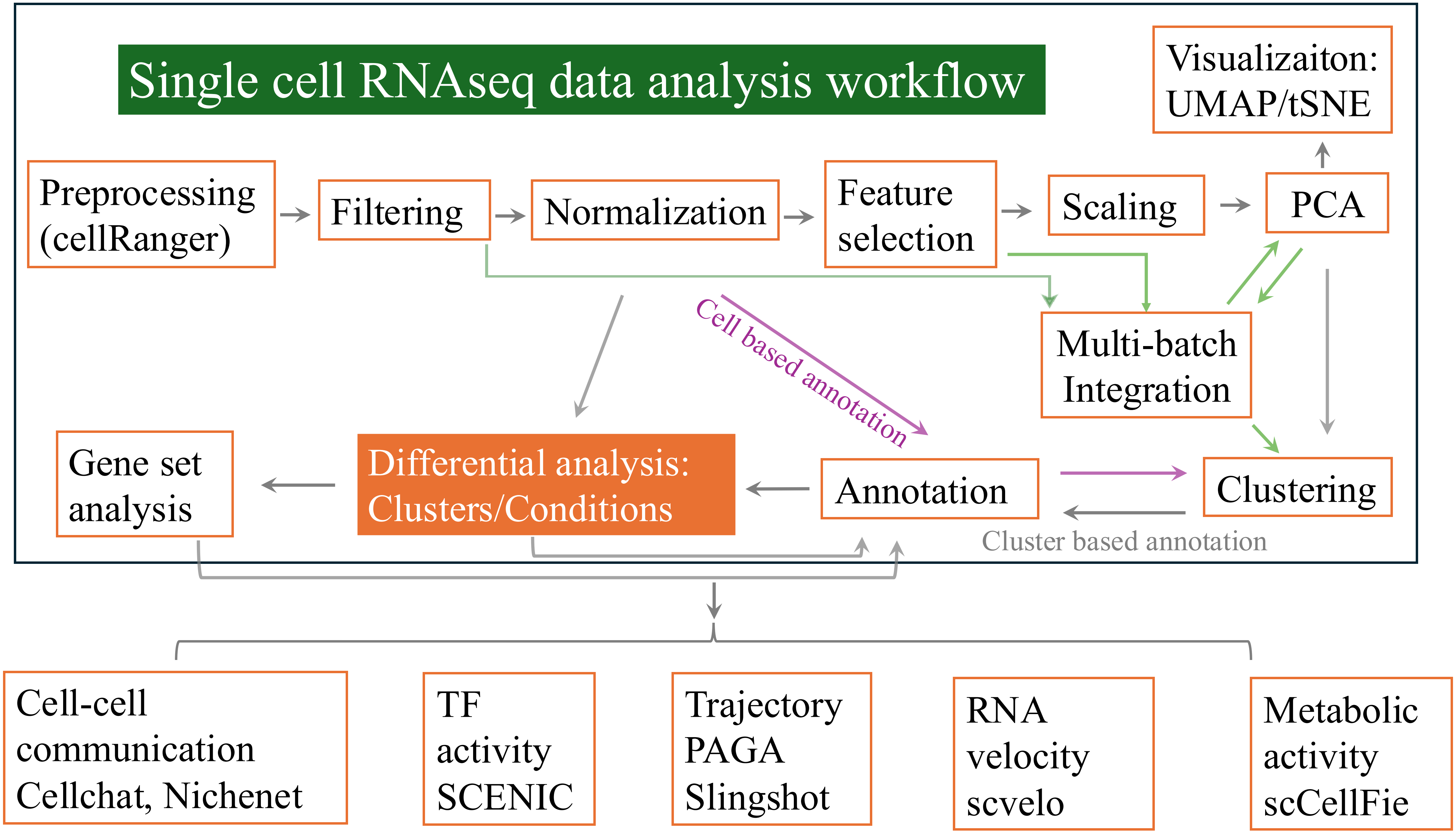


Recommended workflow for annotation



Clarke et al. 2021, Nature protocols

Differential expression analysis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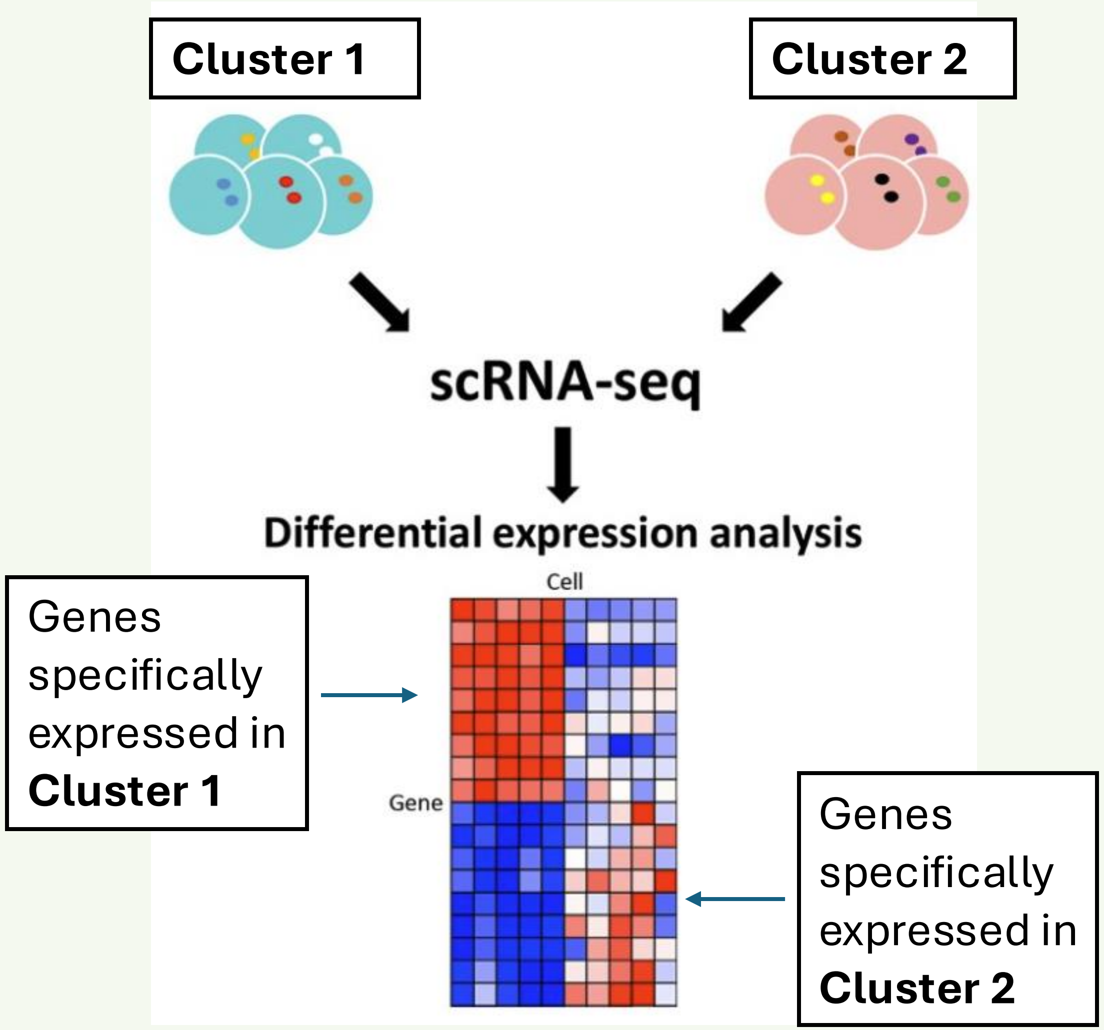
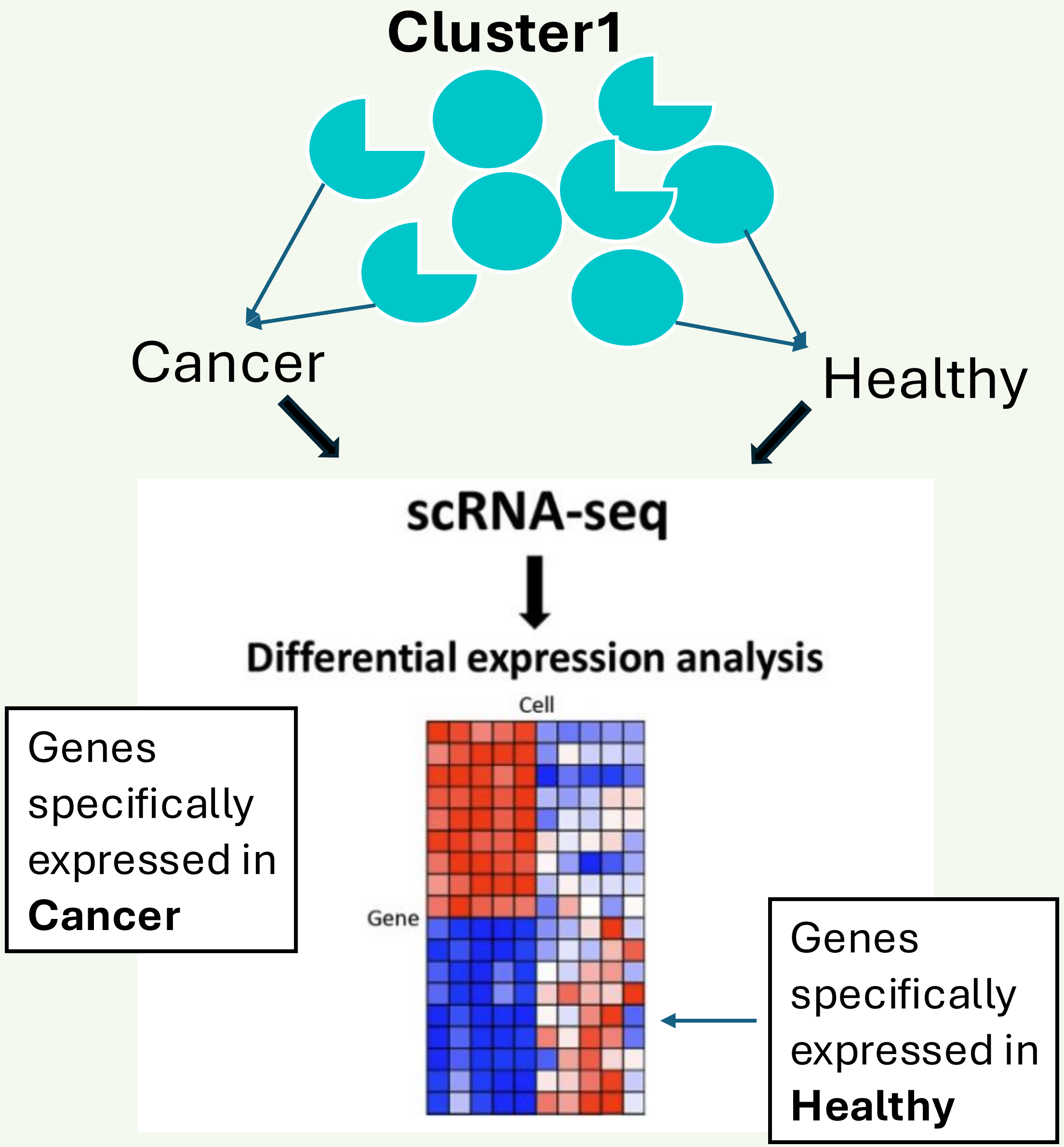
Between clusters



Between conditions



Methods paraticularly developed for scRNAseq data:

Single cell RNAseq data:

- ➢ Count data (possion, NB, Guassian…)
- ➢ Noisy
- ➢ **Zero-inflated** (little material, low seq. depth, sparse, **zero**)
- ➢ Bi-modal
- ➢ Multi-modal

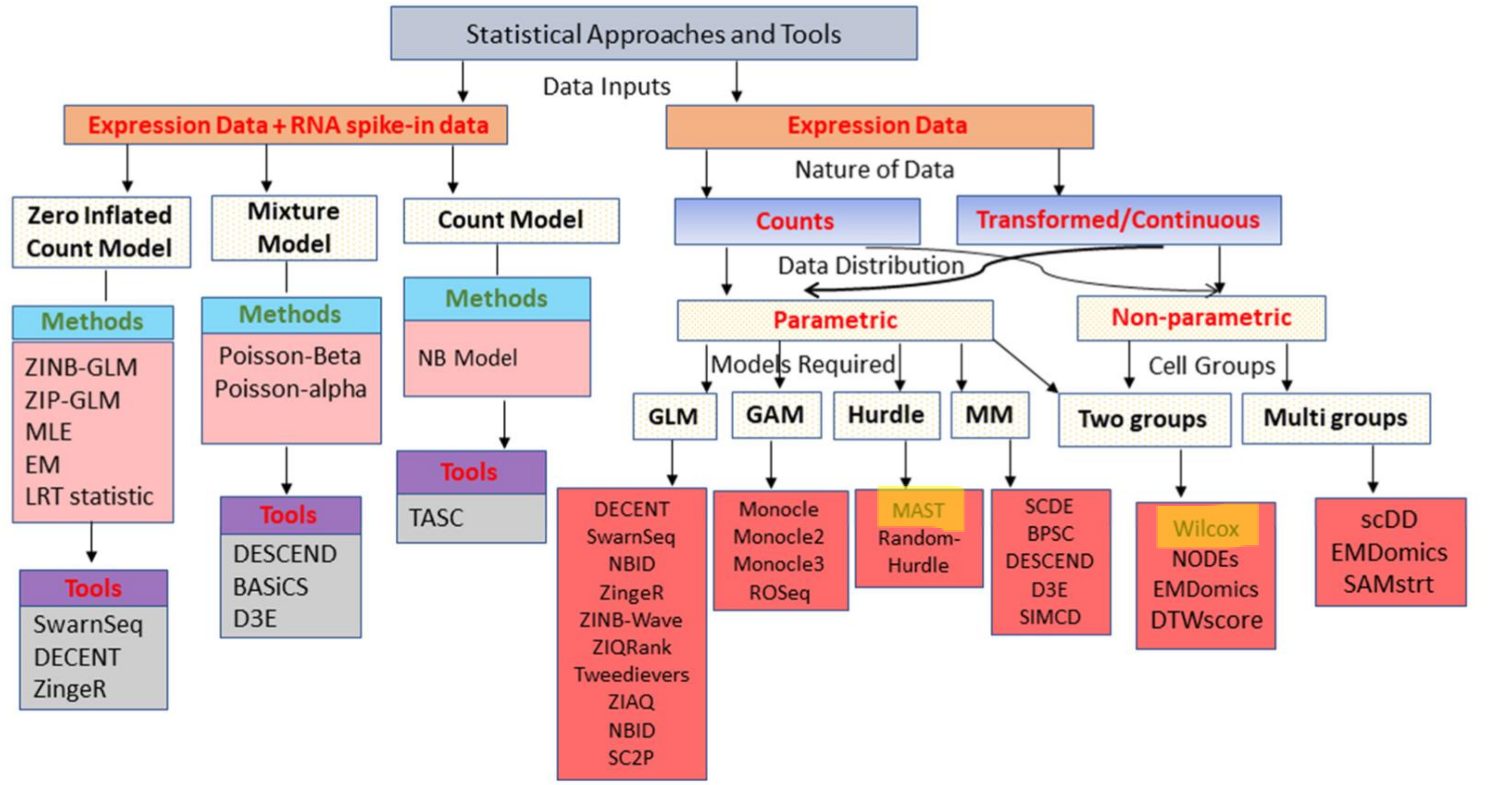
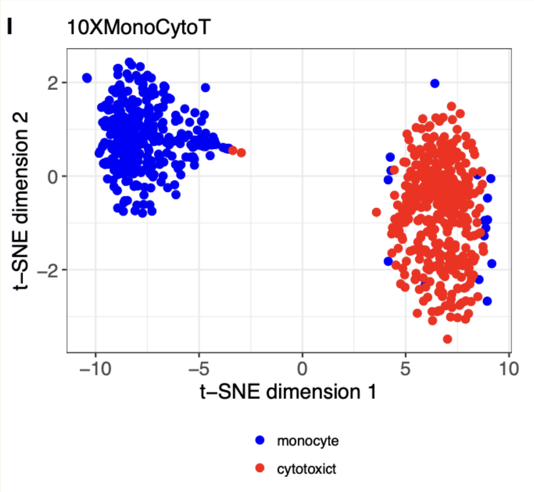
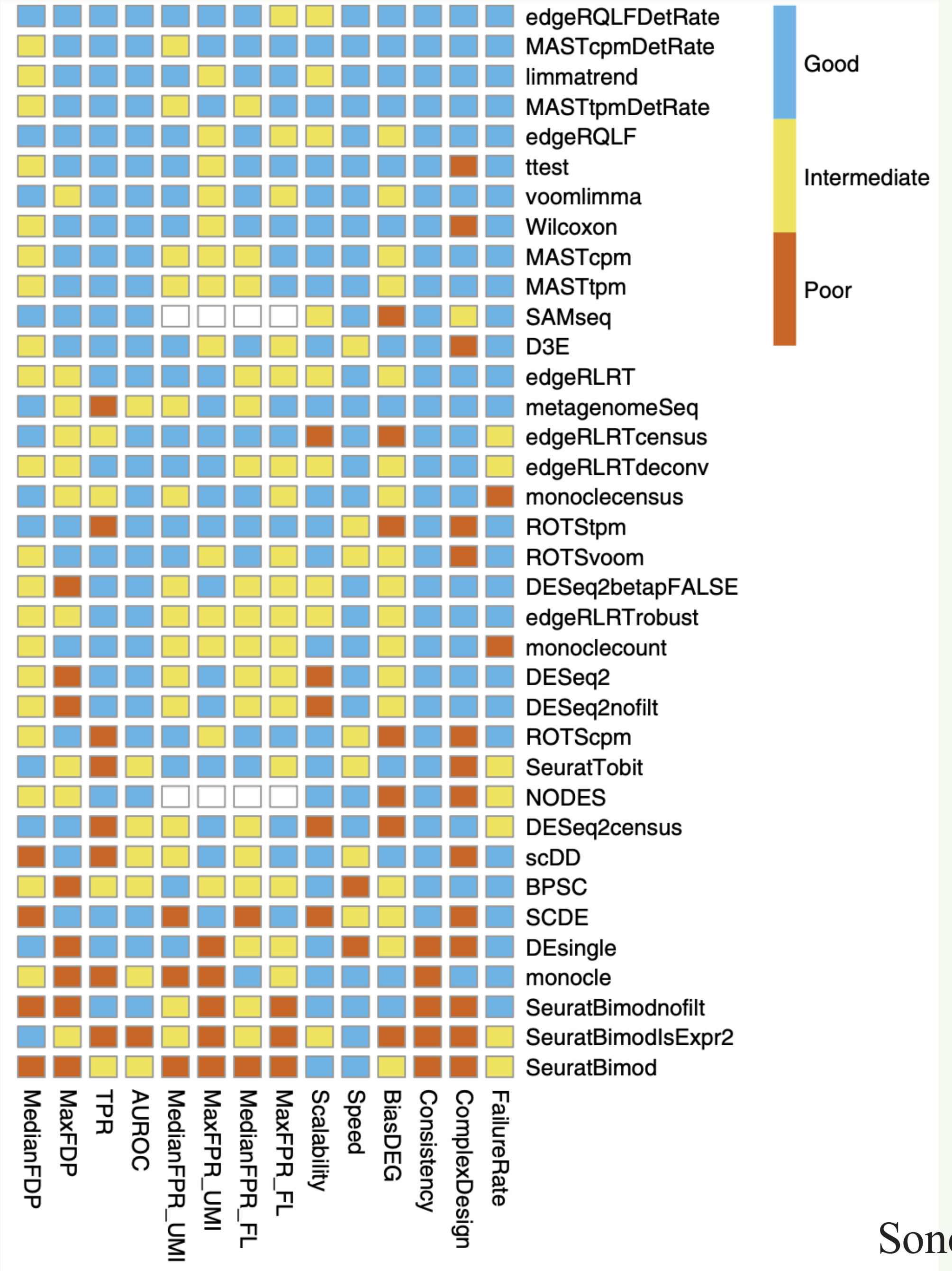


Figure 2. Classification of available statistical approaches and tools used for DEA in single-cell studies. Classification of the approaches is conducted based on the requirement of input data, data

Das et al. 2022 Entropy



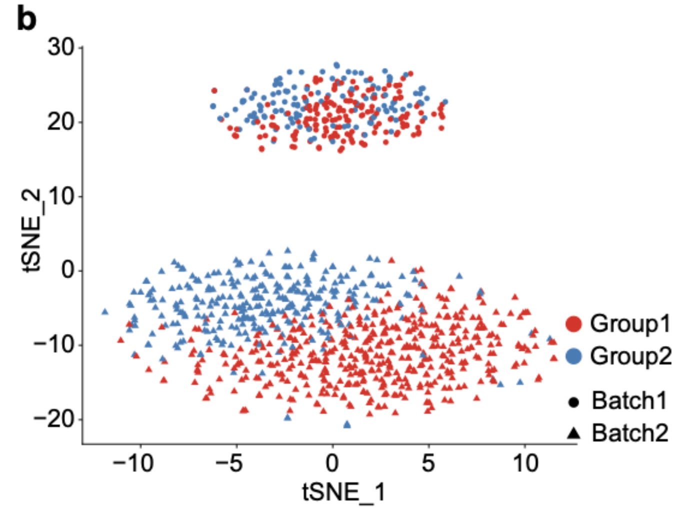


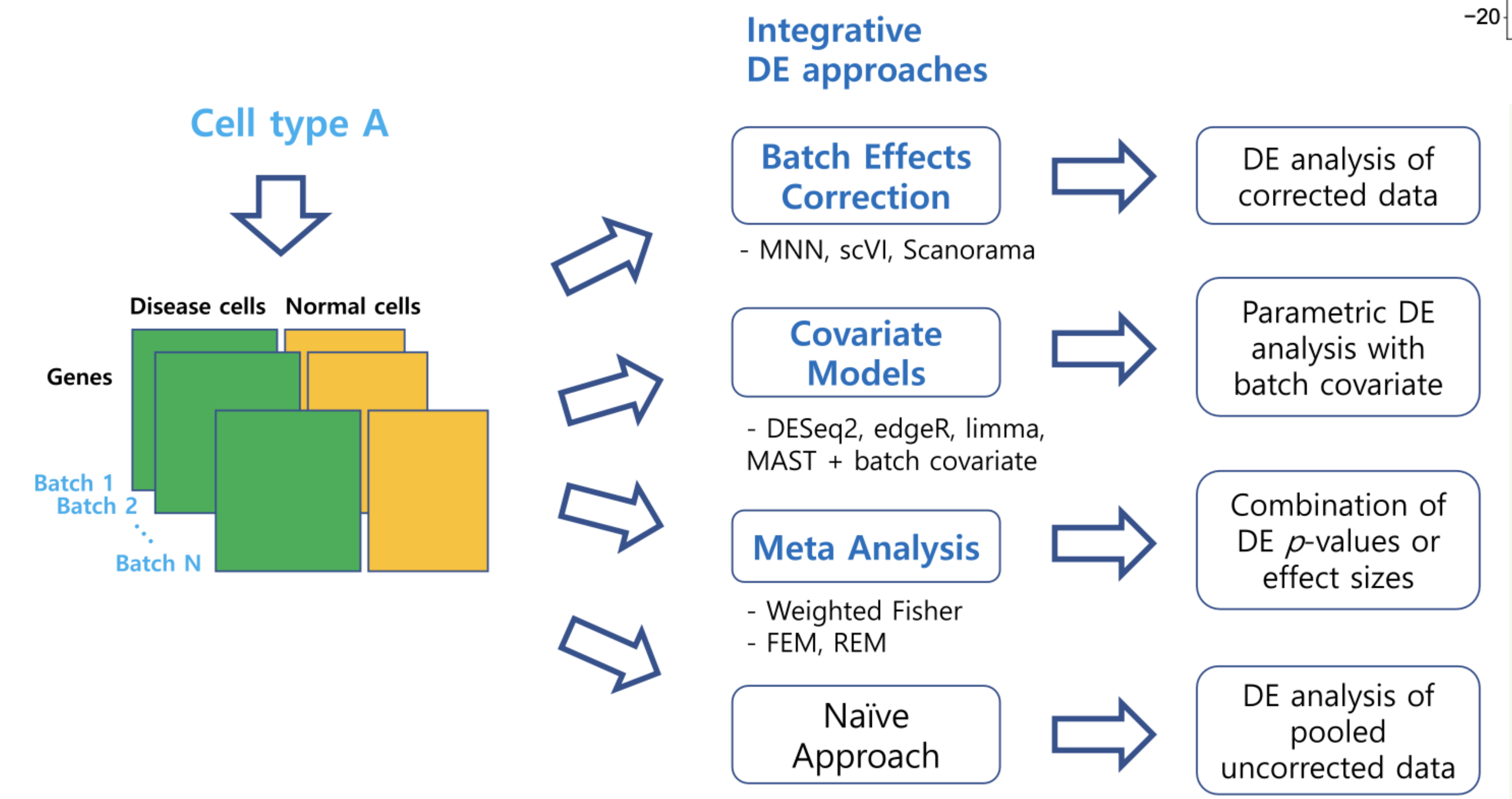
For simple comparison between 2 groups of cells:
No batches…

- ❖ t-test and Wilcoxon work well
- ❖ bulk RNA-seq analysis methods (DESeq2, edgeR, Limma…) do not generally perform worse than those specially developed for scRNA-seq

Soneson&Robinson 2018 *Nat Methods*

When batches added in



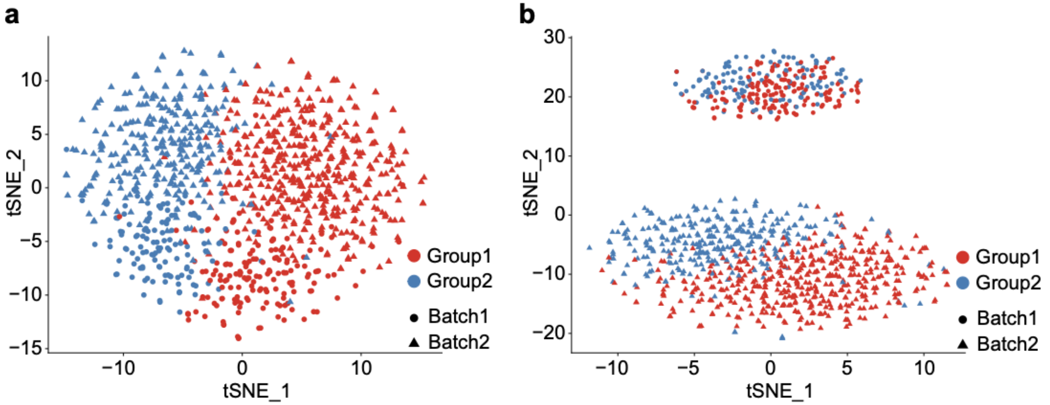


- wilcox

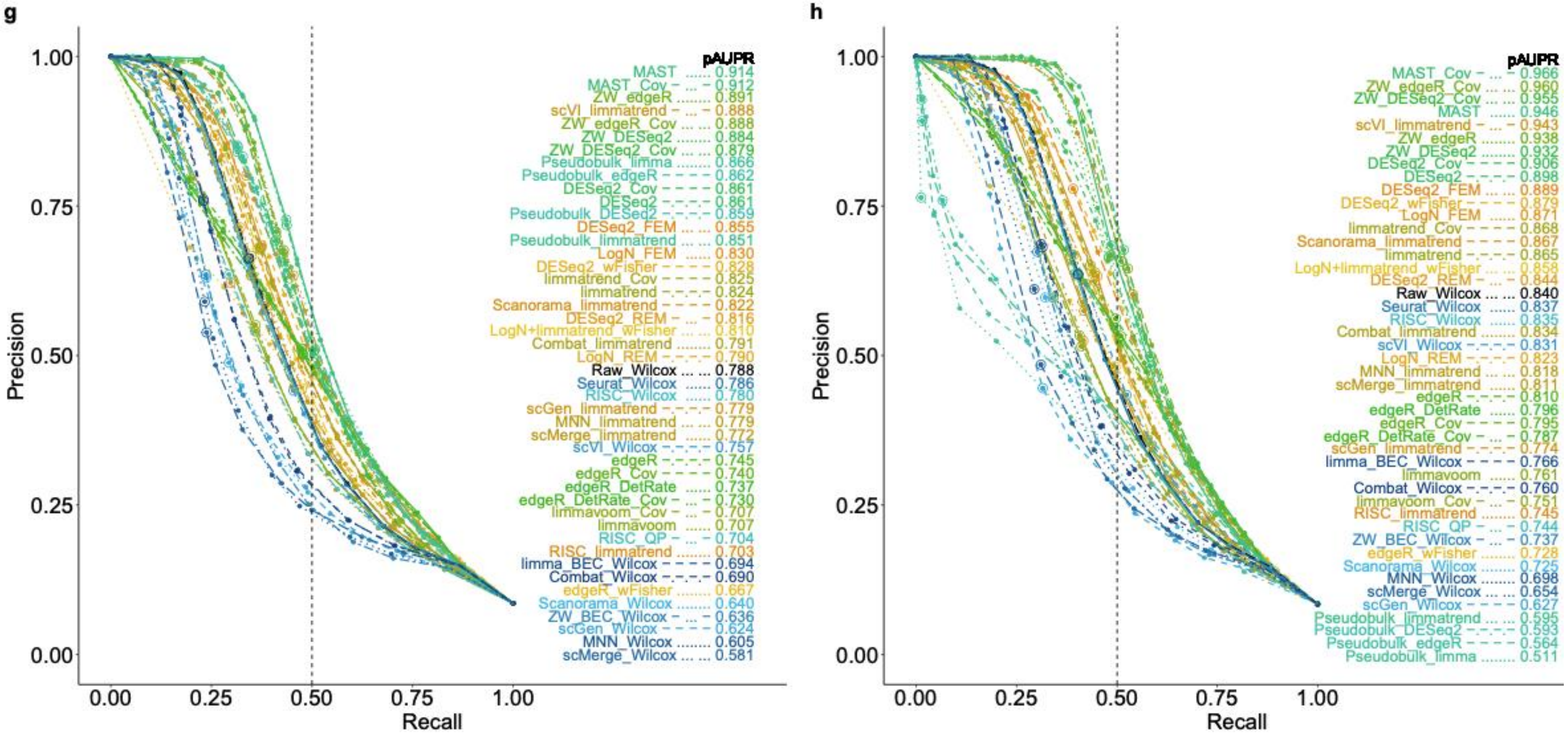
Nguyen et al. 2023 *Nat Commun*

- ❖ Intermediate depth (mean non-zero count of 77)
- ❖ High overall zero rate (> 80%)

Nguyen et al. 2023 *Nat Commun*



Small batch effect **Large batch effect**

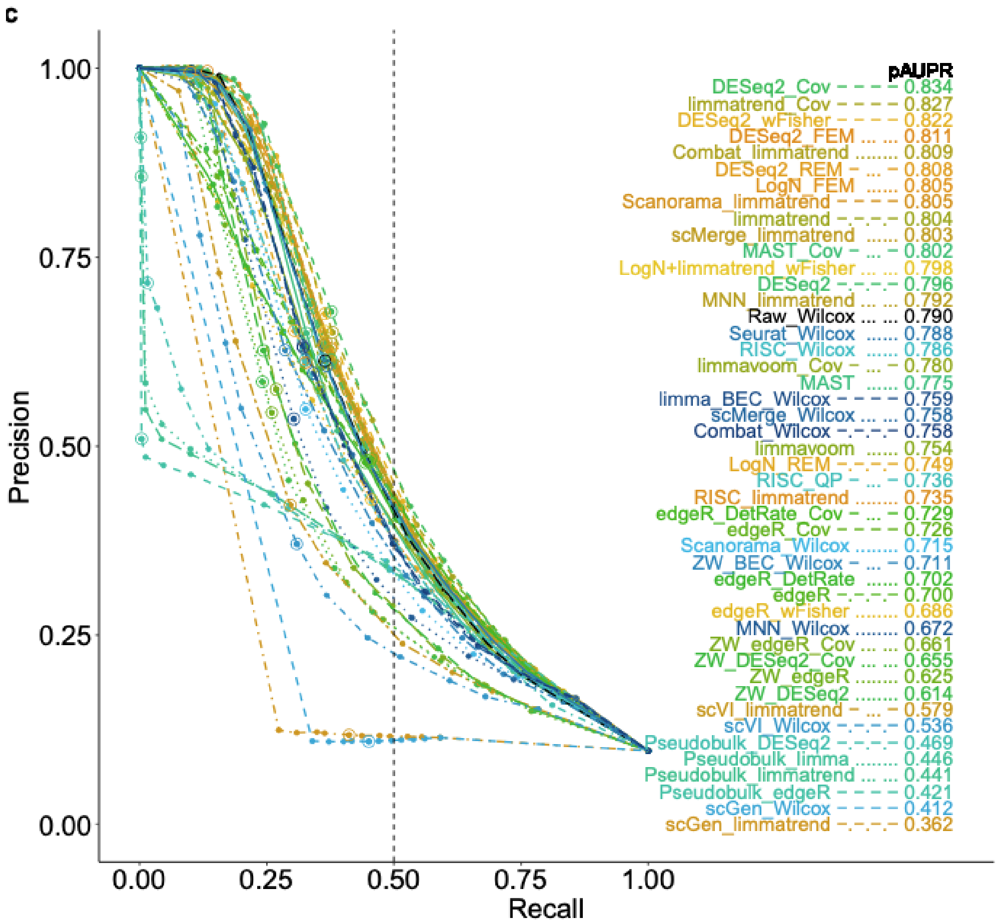


General performance
pAUPR: precision-recall curve {precision=TP/(TP+FP), Recall=TP/(TP+FN)}

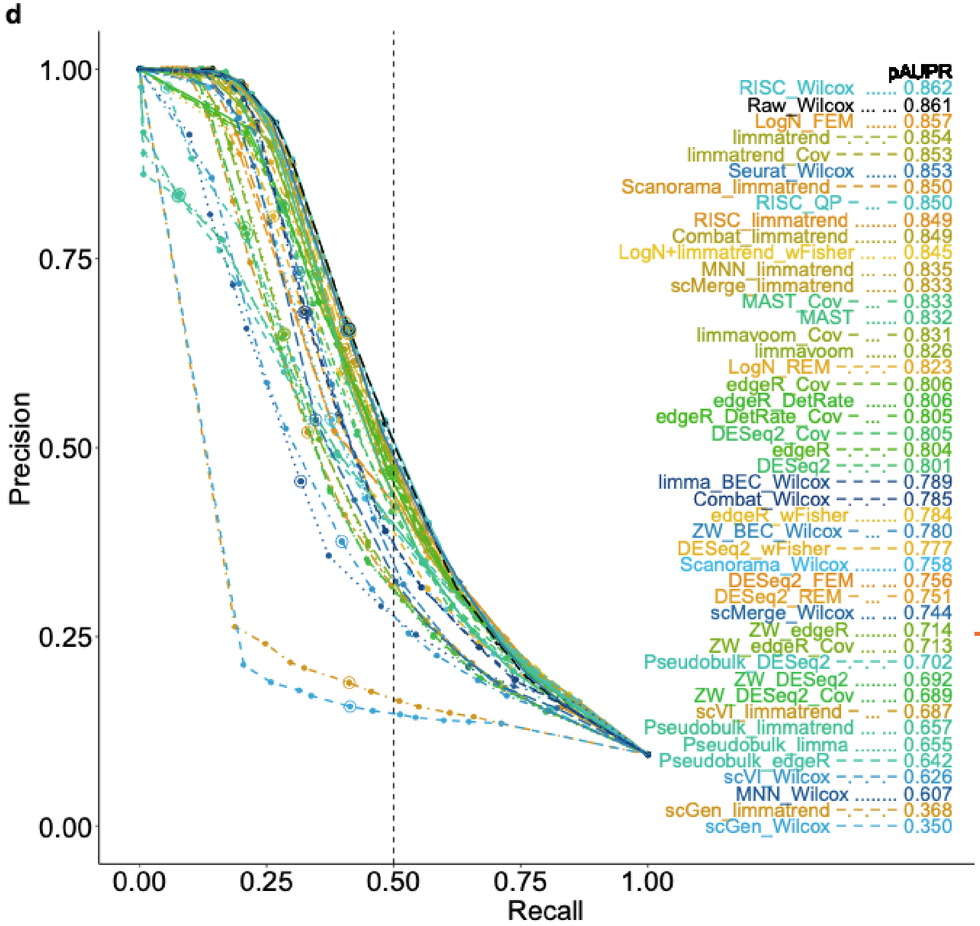
❖ Low depth (mean non-zero count of 10 & 4/10X Genomics)

❖ Zero rate > 80%

Mean non-zero count of 10



Mean non-zero count of 4 (**10X**)



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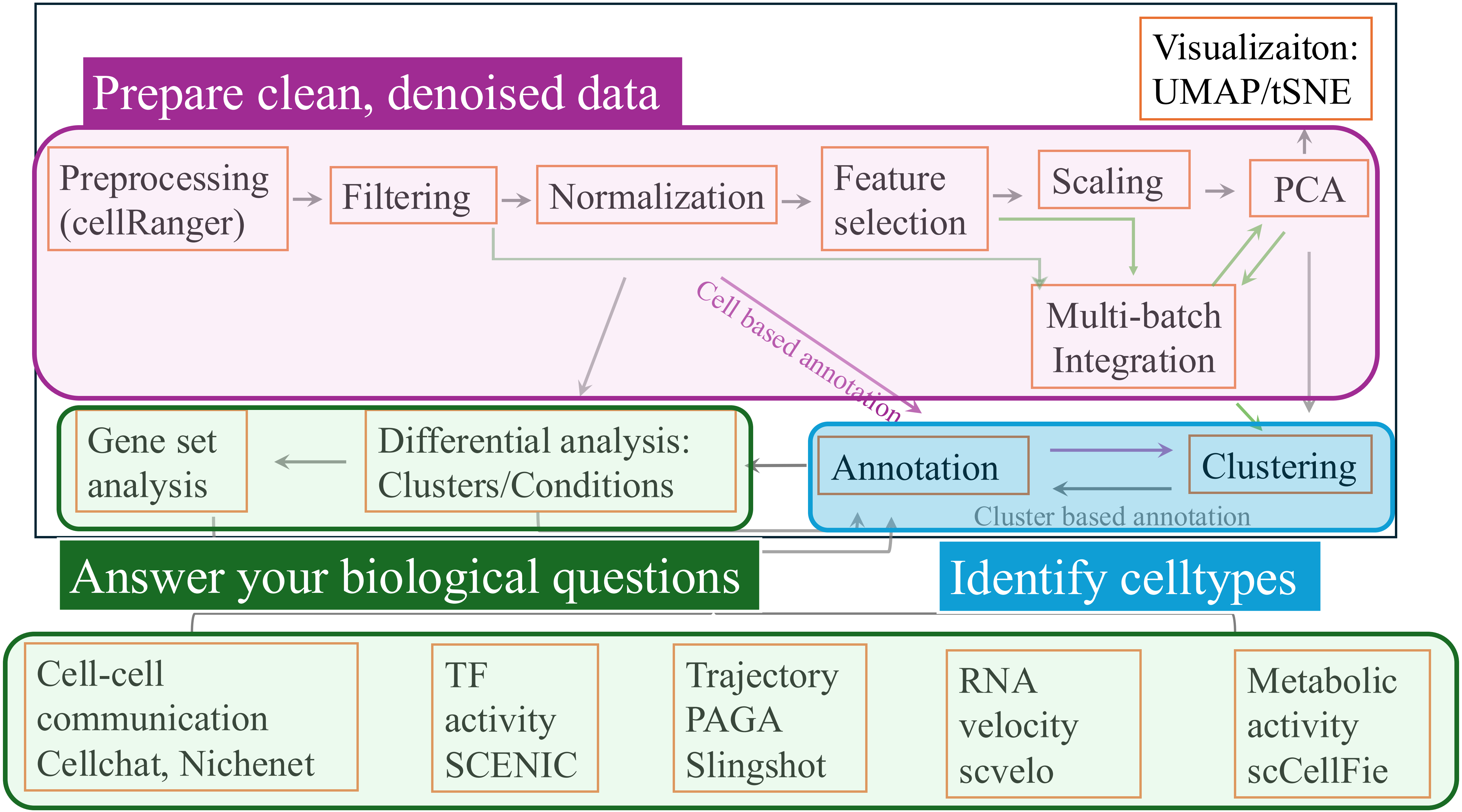
pAUPR: precision-recall curve {precision=TP/(TP+FP), Recall=TP/(TP+FN)}

Nguyen et al. 2023 *Nat Commun*

Basic single cell data analysis summary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References

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- **Cell annotaiton**: Clarke, Zoe A., et al. "Tutorial: guidelines for annotating single-cell transcriptomic maps using automated and manual methods." *Nature protocols* 16.6 (2021): 2749-2764.